

Development of the head and neck

Head development is distinct from that of the trunk, utilizing region-specific genes, signalling mechanisms and morphogenetic processes. This chapter will provide a description of head and neck development based primarily on data from studies on human embryos. Since this information is necessarily purely descriptive, it will be supplemented by observations from lineage and genetic studies in mouse embryos, where the results are likely to be valid for all mammals, including humans. The description of human development is derived from the following sources, which incorporate primary data by the authors, as well as references to older primary source material: Patten (1968), Hamilton et al (1962), O’Rahilly and Müller (1987, 2001), Hinrichsen (1990), Steding (2009). References to experimental studies in the mouse are provided in the text.

Evolution of the vertebrate head was made possible by the origin of a novel cell population, the neural crest. Neural crest cells have the potential to form connective and skeletal tissues in the head, and they make major contributions to the skull. In mammalian embryos, cranial neural crest cells emigrate from the edges of the still-open cranial neural folds, unlike trunk neural crest cells and the cranial crest of other vertebrates, which begin migration only after neural tube closure. Figure 36.1 shows stylized views of human embryos at an early stage of neural crest migration (Fig. 36.1A) and at the end of the crest migration (Fig. 36.1B). It is important to note that these views do not show the neural crest cells themselves; they can be seen at the same developmental stages in mouse embryos in Figure 36.2.

In the mouse, a lineage marker has enabled neural crest cells to be traced from the stage at which they leave the neural epithelium through to their final locations in mature tissues. These studies show three separate populations of neural crest cells migrating from the cranial neural folds; all these give rise to both neuronal and non-neuronal progeny (Jiang et al 2002; see Fig. 36.2). The first of these populations (trigeminal) originates from the diencephalic region of the forebrain, the mid-brain, and rhombomere A of the hindbrain, which subsequently divides to form the first two rhombomeres (see Figs 36.2A–B; see also Fig. 17.2). Those neural crest cells with a neuronal fate contribute to

the trigeminal ganglion. The non-neuronal cells migrate extensively to surround the telencephalon and part of the diencephalon, forming the frontonasal mesenchymal populations; they also migrate lateral to the rhombencephalon to form the mesenchyme of the maxillary and mandibular regions of the first arch (see Fig. 36.2C–D). The second (hyoid) population gives rise to the otic ganglion and then migrates from rhombomere B (which forms rhombomeres 3 and 4), into the second pharyngeal arch. The third (vagal) population has a more extensive origin, from the neural folds caudal to the otocyst, i.e. rhombomere C. These cells will contribute to the ganglia of the glossopharyngeal and vagal nerves, with the non-neuronal vagal crest cells migrating into pharyngeal arches 3, 4 and 6, some of them continuing into the heart to contribute to the division of the cardiac outflow tract (Jiang et al 2000). The neural crest also gives rise to the parasympathetic ganglia and parasympathetic postganglionic nerves in the head and neck. For a full account of the neuronal contribution of neural crest cells, see Chapter 17. In human embryos, histological methods have revealed equivalent cranial neural crest cell origins and migration routes to those of the mouse (O’Rahilly and Müller 2007).

Caudal to the segmented region of the cranial neural tube, neural crest cells from the occipital region (rhombomere D/rhombomere 8) migrate with occipital myoblasts to form the hypoglossal cord, eventually differentiating to form the connective tissue (neural crest) and musculature (myoblasts) of the tongue. No sensory ganglia are formed from the occipital neural crest in human embryos. The origin of the rhombomeres and brain regions in human embryos is shown in Figures 17.2 and 17.12.

Ectodermal placodes appear as patches of columnar or pseudostratified epithelium within the otherwise squamous epithelium of the surface ectoderm. Three pairs of placodes contribute to sense organs, forming the olfactory epithelium, the lens and the otocyst, undergoing morphogenesis to form a pit and then (for the lens and otic pits) a closed cyst (vesicle). Epipharyngeal placodes are situated within the dorsal ectoderm of pharyngeal arches 2, 3 and 4; some of their cells

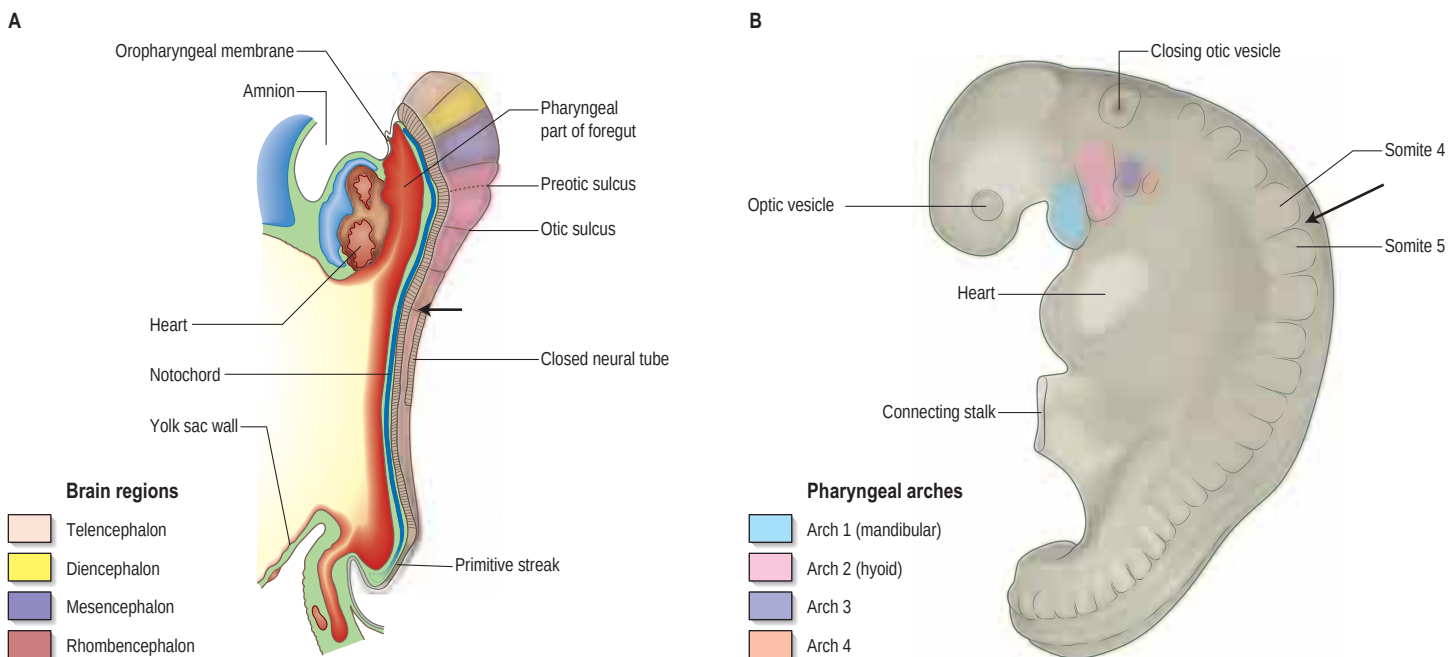


Fig. 36.1 Human embryos during the early stages of head development. The occipital–cervical boundary is indicated by an arrow. **A**, A sagittal section of a stage 10 embryo, in which neural tube closure has begun in the future occipito–cervical region. **B**, A lateral view of a stage 12 embryo; the occipital–cervical boundary between somites 4 and 5 is indicated by an arrow (see also Fig. 17.2).

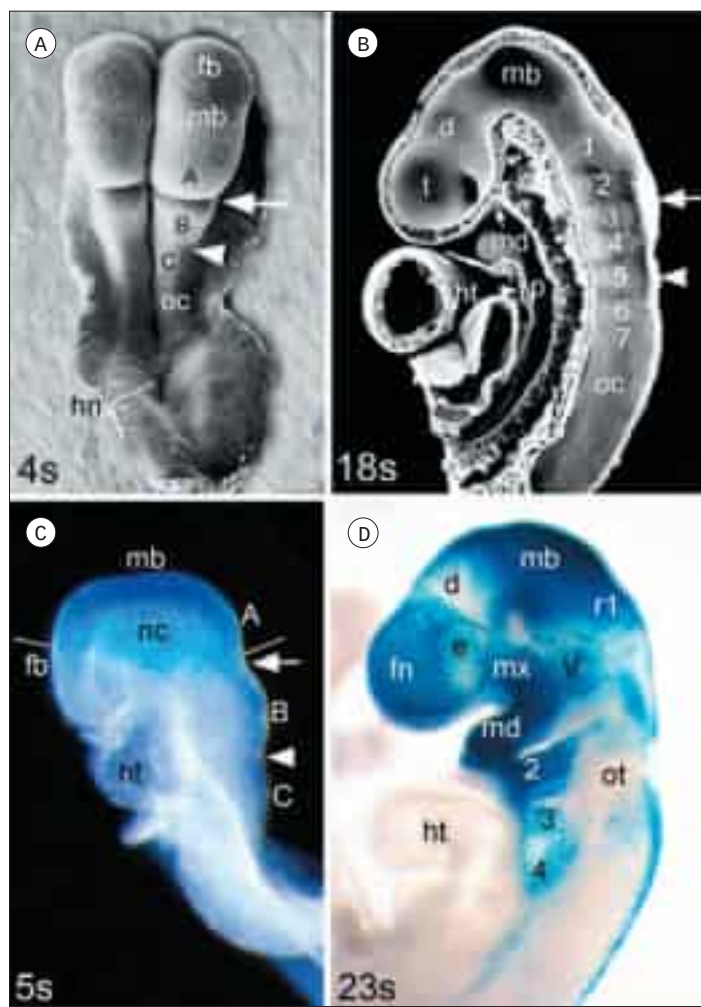


Fig. 36.2 Mouse embryo heads during the period of cranial neural crest cell migration. **A–B**, Scanning electron micrographs (**A**, dorsal view; **B**, median section) of embryos near the start (4 somite pairs) and end (18 somite pairs, interior view) of the period of neural crest cell migration. At the 4-somite stage, the neural folds are convex and the hindbrain is divided into prorhombomeres A, B and C by the preotic (arrow) and otic (arrowhead) sulci; at the 18-somite stage, the neural tube is closed and the hindbrain is divided into seven rhombomeres (numbered) and the unsegmented occipital region (oc). Rathke's pouch is indicated by a small arrow on **B**. **C–D**, Lateral views of embryos at the 5- and 23-somite stage, respectively, carrying a permanent marker for neural crest cells and the neural epithelium from which the trigeminal crest cells originate (blue stain). Trigeminal neural crest cells (nc) migrate as a veil-like mass beneath the surface ectoderm, originating as a continuous group from prorhombomere A, the midbrain and the caudal diencephalic region (d) of the forebrain, into the frontonasal region (fn) and the first arch (white lines indicate the level of the migrating edge). After migration, frontonasal neural crest cells (fn) cover the telencephalon and rostral diencephalon, surrounding the eye (e); maxillary (mx) and mandibular (md) neural crest populates the first pharyngeal arch; the trigeminal ganglion primordium (V) is also present. Hyoid neural crest cells from prorhombomere B have populated the hyoid arch and the primordium of ganglion VII/VIII, rostral to the otocyst (ot); vagal crest cells from prorhombomere C are migrating into arches 3 and 4, and starting to form the primordia of the glossopharyngeal and vagal ganglia, caudal to the otocyst. Other abbreviations: d, diencephalon; hn, Hensen's node; ht, heart; mb, midbrain; p, pharynx; r1, rhombomere 1 of the hindbrain; t, telencephalon. (**A** and **B**, With permission from Morriss-Kay GM, Wilkie AOMW 2005 Growth of the normal skull vault and its alteration in craniosynostosis. *J Anat* 207:637–53. **C** and **D**, With permission from Jiang X, Iseki S, Maxson RE et al 2002 Tissue origins and interactions in the mammalian skull vault. *Dev Biol* 241:106–16.)

undergo epithelial to mesenchymal transformation, joining the underlying neural crest cell condensations to form sensory neuroblasts in the cranial sensory ganglia V, VII, IX and X. Some cells from the otocyst and an adjacent placode similarly delaminate to contribute to the ganglion of cranial nerve VIII (see Fig. 17.18).

The epithelial to mesenchymal transformation of some cells from ectodermal placodes contributes a third component to the cranial mesenchyme. It is important to note that the term mesenchyme simply

describes a loose cellular organization with a rich extracellular matrix, in contrast to the sheets of simple epithelial tissue that form the outer ectodermal and inner endodermal layers of early embryos. Prior to the migration of neural crest cells from the edges of the cranial neural folds to their final destinations, the only mesenchyme in the embryonic head is 'primary mesenchyme', i.e. mesenchymal cells formed by epithelial–mesenchymal transformation in the primitive streak. There is no lateral plate mesenchyme in the head; all of the cranial mesenchyme is paraxial. Cranial mesenchyme derived from placodal, primary mesenchymal and neural crest sources has specific distributions and specific derivatives, and is not interchangeable.

The first transverse division to form in the early embryonic brain is the preotic sulcus (see Figs 36.1A, 36.2A), which defines the boundary between prorhombomeres A and B. This early division is of major functional and organizational significance, since it separates the regions of origin of the skeletogenic neural crest populations that make a major contribution to the skull from those that form skeletal structures in the neck (hyoid bone and larynx). During cranial neurulation, the prorhombomeres undergo subdivision so that, by the time the brain region of the neural tube closes, seven rhombomeres can be clearly distinguished; the occipital region (adjacent to the four occipital somites) remains unsegmented (see Fig. 36.2B). (By analogy with the segmented rhombencephalon, it is also referred to as prorhombomere D/rhombomere 8.) The segmental organization of the embryonic head caudal to prorhombomere A is related to the expression of evolutionarily conserved *HOX* genes (written *Hox* in the mouse) that have their rostral boundaries at specific rhombomere divisions; they are also expressed in the corresponding neural crest cells (reviewed by Santagati and Rijli 2003; see also Fig. 17.16). Genetic experiments in the mouse and other vertebrates have shown that skeletal patterning of the first pharyngeal arch depends on the absence of *Hox* gene expression in the neural crest cells migrating into it; skeletal patterning of the second arch depends on the expression of *Hoxa2*. Investigation of *Hox* gene expression in arches 3–6 has not revealed clear skeletal patterning functions.

EMBRYONIC PHARYNX AND PHARYNGEAL ARCHES

The most cranial portion of the foregut, the embryonic pharynx, is the scaffolding around which the face, palate and anterior neck structures are built. The development of this region from neural crest, paraxial mesenchyme, surface ectoderm and foregut endoderm involves spatio-temporal coordination of cell movement, tissue growth and tissue interactions. As successive populations of neural crest cells migrate around the pharynx at progressively more caudal levels, five pairs of pharyngeal arches are formed (numbered 1, 2, 3, 4 and 6 for comparative anatomy reasons). This process is complete by stages 14–15 (5 weeks). Pharyngeal clefts (grooves) separate the arches externally; they are matched internally by internal depressions, the pharyngeal pouches. Pharyngeal arches are also known as branchial arches because of their evolutionary origin supporting the gills in the earliest vertebrates. Many of the changes seen during development of the mammalian pharynx reflect the functional evolutionary origins of this region. Since the term 'branchial' means 'of the gills', the term 'pharyngeal arch' will be used in the following account of human embryology.

Each pharyngeal arch consists of an epithelial covering, ectoderm externally and endoderm internally, filled with mesenchyme that is mainly of neural crest origin, with a contribution from primitive streak-derived mesenchyme (paraxial mesenchyme) (Fig. 36.3B; Table 36.1). The neural crest cells of each arch form a skeletal element and associated connective tissue; they also give rise to the walls of an aortic arch blood vessel, which is lined with endothelium derived from angiogenic mesoderm. Paraxial mesenchyme also forms the muscle associated with each pharyngeal arch. In mouse embryos, a cell lineage marker for mesoderm has shown that, by embryonic day 10.5 (equivalent to human stage 13, or about 32 days), mesoderm-derived mesenchymal cells have contributed both angiogenic and myogenic cells to the pharyngeal arches (Yoshida et al 2008). Motor and sensory innervation derives from arch-specific cranial nerves. For an overview of the organization of these tissues in the pharynx, see Figure 12.4; for their fate, see Table 36.1 and Figures 36.5–36.7. Since the organization of the embryonic pharynx and related structures is similar in all vertebrate embryos, this phase has been termed 'pharyngula'.

The first pharyngeal arch, unlike the other arches, possesses dorsal and ventral processes, and appears C-shaped in lateral view. It is also termed the 'mandibular arch'; however, this is not entirely appropriate because its proximal (dorsal) end also contributes to maxillofacial and palatopharyngeal structures and to the middle ear. The second

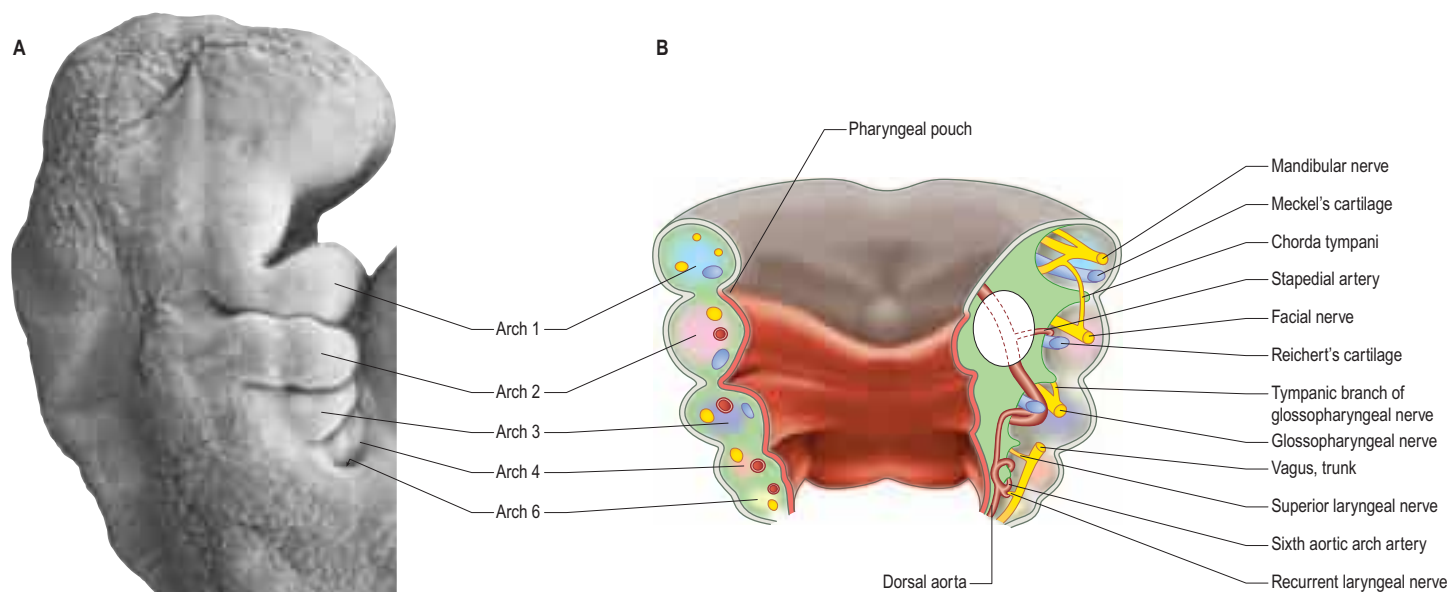


Fig. 36.3 Pharyngeal arch development. **A**, A scanning electron micrograph of a human embryo (stage 14/15), showing the right lateral aspect of the pharyngeal arches; auricular hillocks are visible on arch two. **B**, The pharyngeal region (stage 15) viewed from the dorsal aspect. The whole left side and part of the right side of the endodermal roof of the pharynx has been removed to show the grooves in its lateral walls and floor, and the nerves, arteries and cartilages of the arches. The position of the right otocyst is shown as a clear oval. (A, Courtesy of Prof G Steding, Georg-August-University of Göttingen, Göttingen, Germany.)

Table 36.1 Derivatives of the pharyngeal arches

Arch number	Arch name	Embryonic cartilage	Cartilage derivative	Muscle	Nerve	Artery
1	Mandibular	Quadrate Meckel's	Incus Malleus Anterior ligament of malleus Spine of sphenoid Sphenomandibular ligament Genial tubercle of mandible	Tensor tympani Muscles of mastication Mylohyoid Anterior belly of digastric Tensor veli palatini	Trigeminal (V) Mandibular division	First aortic arch artery [transitory]
2	Hyoid	Reichert's	Stapes Styloid process of temporal bone Stylohyoid ligament Lesser horn and upper part of body of hyoid bone	Stapedius Stylohyoid Facial muscles, incl. Buccinator Platysma Posterior belly digastric	Facial (VII)	Stapedial artery [transitory]
3	Third		Greater horn and lower part of body of hyoid bone	Stylopharyngeus	Glossopharyngeal (IX)	Common carotid artery, first part of internal carotid artery
4	Fourth		Thyroid cartilage Corniculate cartilage Cuneiform cartilage	Pharyngeal and extrinsic laryngeal muscles, levator veli palatini	Vagus (X) Pharyngeal branch	Proximal part of subclavian artery on the right Arch of aorta between origins of left common carotid and left subclavian arteries
6	Sixth		Arytenoid cartilages	Intrinsic laryngeal muscles	Vagus (X) Recurrent laryngeal branch	Part between the pulmonary trunk and dorsal aorta becomes ductus arteriosus on left, disappears on right

pharyngeal arch is termed the hyoid arch because of its contribution to the hyoid bone (and to other parts of the hyomandibular apparatus in fishes). The third, fourth and sixth arches are not named.

The mandibular arches, first seen at stage 10, grow ventromedially in the floor of the pharynx to meet in the median plane, forming the ventral border of the early mouth above the cardiac (pericardial) prominence. The maxillary processes are not clearly visible as surface structures until stage 13; their enlargement coincides with the proliferation of the frontonasal mesenchyme to form the nasal swellings. The hyoid arches appear at stage 11, the third arches at stage 12 and the fourth arches by stage 13. The sixth arch can only be identified by the arrangement of the arch arteries and nerves, and by a slight projection on the pharyngeal aspect. The floor of the pharynx is formed mainly by ventral apposition of the arches. The inferior border of the second arch grows over the lower arches and encloses an ectodermal depression termed the cervical sinus (Figs 36.3A, 36.4).

SKELETAL ELEMENTS OF THE PHARYNGEAL ARCHES

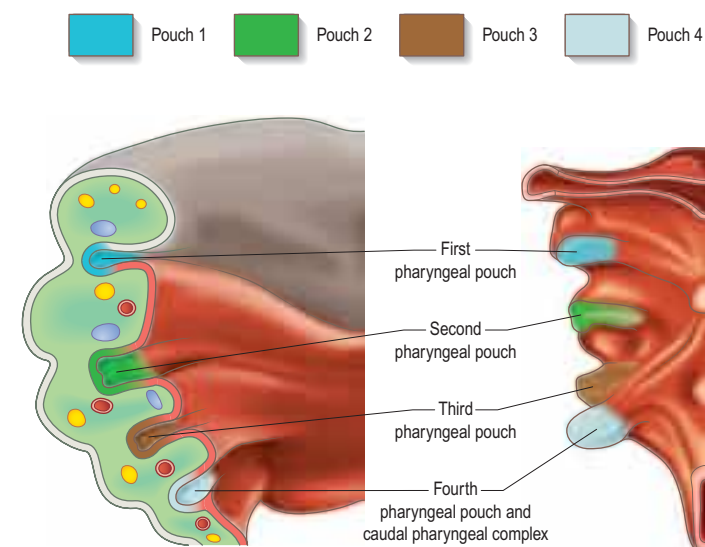
The skeletal and connective tissue elements of arches 1–3 are formed by condensation of neural crest-derived mesenchyme, which subsequently chondrifies in part or all of its length. Where chondrogenesis

is complete (first and second arch cartilages), the element extends dorsally until it comes into contact with the skull base lateral to the hind-brain (Fig. 36.5). An arch cartilage may remain as cartilage, undergo endochondral ossification or become ligamentous, or a combination of these (Fig. 36.6).

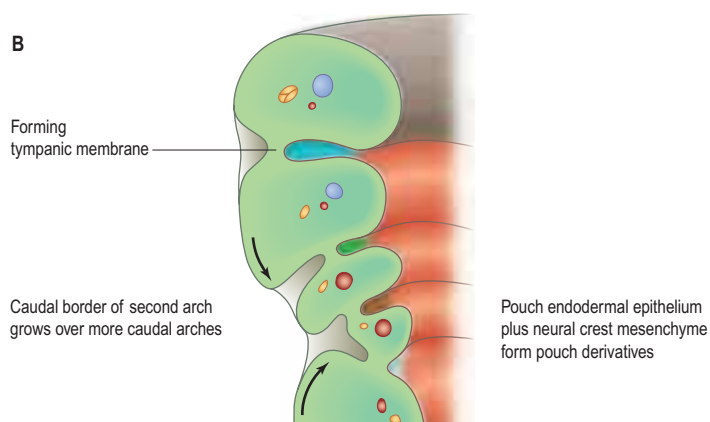
The first arch cartilage, Meckel's cartilage, forms the embryonic lower jaw skeleton. After the mandible forms by intramembranous ossification of the neural crest-derived mesenchyme lateral to it, Meckel's cartilage degenerates but its sheath persists as the anterior malleolar and sphenomandibular ligaments. At its proximal end, endochondral ossification forms two of the middle ear bones, the incus and malleus. These are homologous with the quadrate and articular bones of reptiles, and the joint between them is evolutionarily derived from the reptilian jaw articulation. In reptiles, the quadrate forms from the caudal end of the palatopterygoquadrate cartilage; the mammalian evolutionary derivatives of the palatopterygoquadrate are thought to include part of the greater wing of the sphenoid bone and the roots of its pterygoid plates, in addition to the incus. The portion of Meckel's cartilage that extends from the mental foramen almost to the site of the future mandibular symphysis probably becomes ossified and incorporated into the intramembranous mandibular bone; the remainder of the cartilage is ultimately absorbed.

The second arch cartilage, Reichert's cartilage, extends from the proximity of the otic capsule dorsally to the median plane ventrally. It

A Pharyngeal pouches



B



C

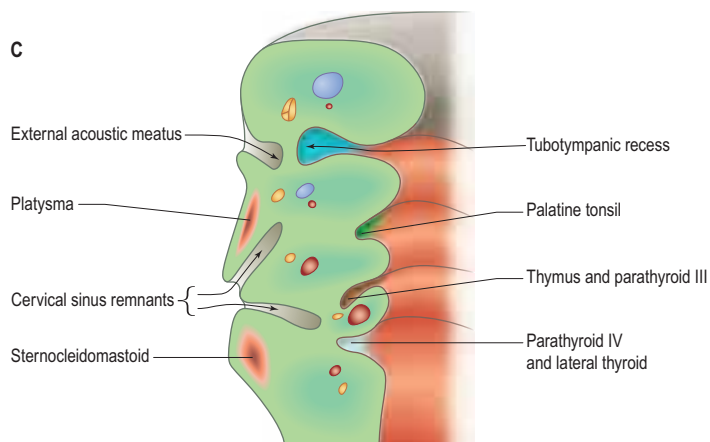


Fig. 36.4 Pharyngeal pouch development. **A**, The arrangement of the early pharyngeal pouches: on the left, the internal aspect of the pharyngeal floor viewed from above; on the right, the external aspect of the pharyngeal floor viewed from below. **B**, A coronal section of the left side of the pharynx at stage 18, showing changes to the pharyngeal pouches internally and pharyngeal clefts externally. **C**, A coronal section of the left side of the pharynx at stage 19. See also [Figure 36.18](#) for further development.

ossifies to form the upper part of the body and lesser cornua of the hyoid bone distally, and the styloid process proximally; between these two bones, it becomes ligamentous to form the stylohyoid ligament. It also gives rise to the most ancient of the three auditory ossicles, the stapes (homologous with the rod-shaped columella auris of reptiles and birds). The cartilages of the third, fourth and sixth arches form only anterior neck skeletal structures: the lower part of the body and greater cornua of the hyoid bone (third arch), and the laryngeal cartilages. Neural crest cells also give rise to ligaments, tendons and connective tissue in these arches. However, in contrast to the NC-derived cartilages

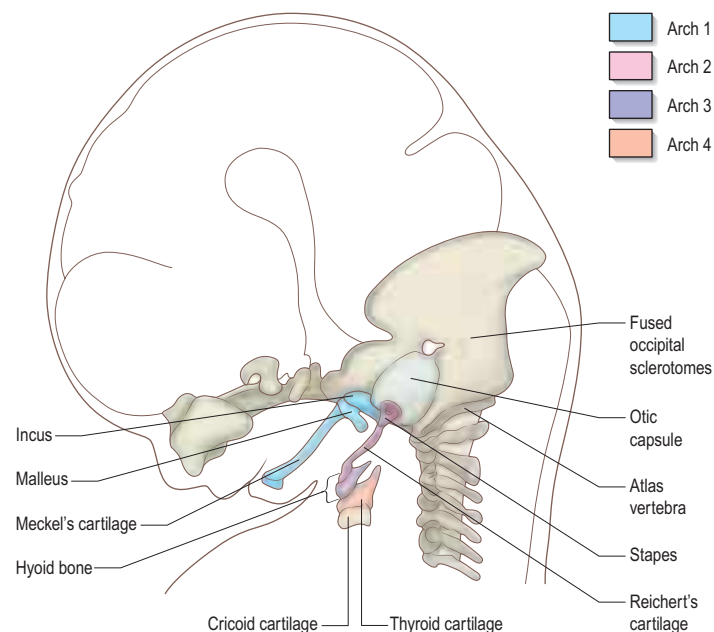


Fig. 36.5 The chondrocranium and arch cartilage derivatives, 7 weeks (stage 20).

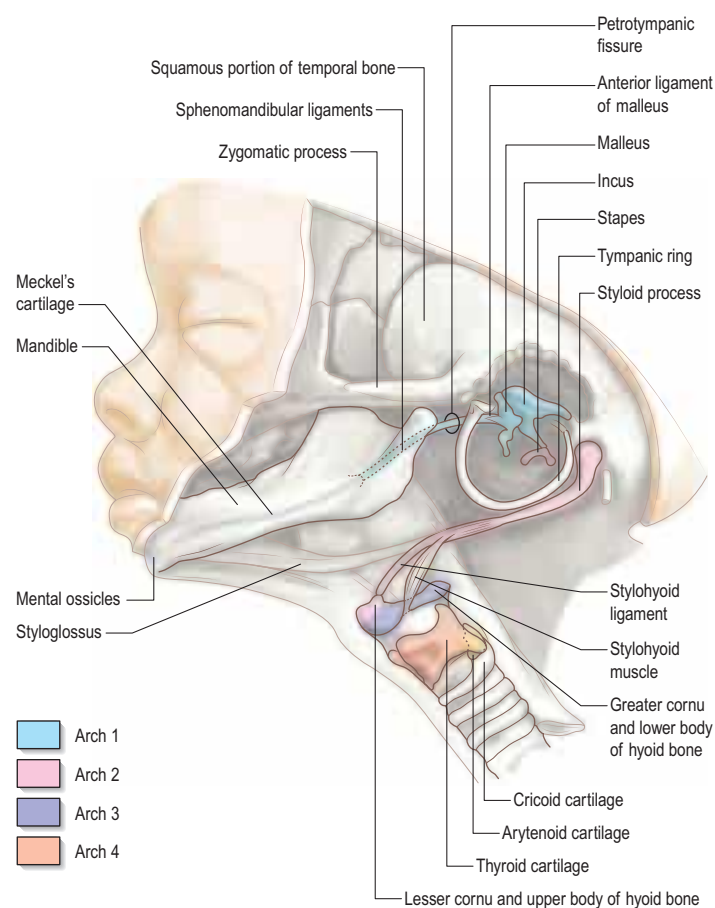


Fig. 36.6 Derivatives of the pharyngeal arch cartilages in a 4–5-month fetus.

of arches 1–3, the laryngeal cartilages associated with arches 4 and 6 (thyroid, cricoid and arytenoid) are mesodermal in origin.

MUSCLES OF THE PHARYNGEAL ARCHES

The striated muscle of each arch (sometimes termed branchiomeric muscle) is derived from the cranial paraxial mesenchyme. Rostral to the occipital region, the paraxial mesenchyme of the head is unsegmented, and the suggestion that a segmental pattern of seven cranial somitomeres exists is not generally accepted today. In the occipital region of the head, epithelialization takes place to form four pairs of somites;

these are similar to those of the trunk, except that the first pair is rather poorly defined. Myoblasts migrate from the paraxial mesenchyme to sites of future muscle differentiation and form pre-muscle condensations prior to the development of any skeletal elements. The pattern of primary myotube alignment for any one muscle is specified by the surrounding neural crest-derived mesenchyme and is not related to the source of the myoblasts. The rate and pattern of muscle maturation are closely associated with the development of the skeletal elements but remain unattached until an appropriate time. Table 36.1 and Fig. 36.7 indicate the muscle masses of each arch, their innervation and their derivatives in the adult.

The muscle mass of the mandibular part of the first arch forms tensor tympani, tensor veli palatini, mylohyoid, anterior belly of digastric, and the masticatory muscles (see Fig. 36.7 and Table 36.1). Tensor tympani retains its connection with the skeletal element of the arch through its attachments to the malleus, and tensor veli palatini remains attached

to the base of the medial pterygoid process, which may be derived from the dorsal cartilage of the first arch. All of these muscles are supplied by the mandibular nerve, the mixed nerve of the first arch. The maxillary division of the trigeminal nerve has no motor component, suggesting that no muscles are derived from maxillary mesenchyme.

The muscles of the second arch migrate widely but retain their original nerve supply from the facial nerve; migration is facilitated by the early obliteration of part of the first pharyngeal cleft and pouch. Stapedius, stylohyoid and posterior belly of digastric remain attached to the hyoid skeleton but the facial musculature, platysma, auricular muscles and epicranium all lose connection with it.

The muscle masses from the third and fourth arches are adapted to form the musculature of the pharynx, larynx and soft palate. Stylopharyngeus is a third arch muscle, cricothyroid develops in the fourth arch, and the other laryngeal muscles are derived from the sixth arch. The precise origin of the remaining palatal muscles and the pharyngeal

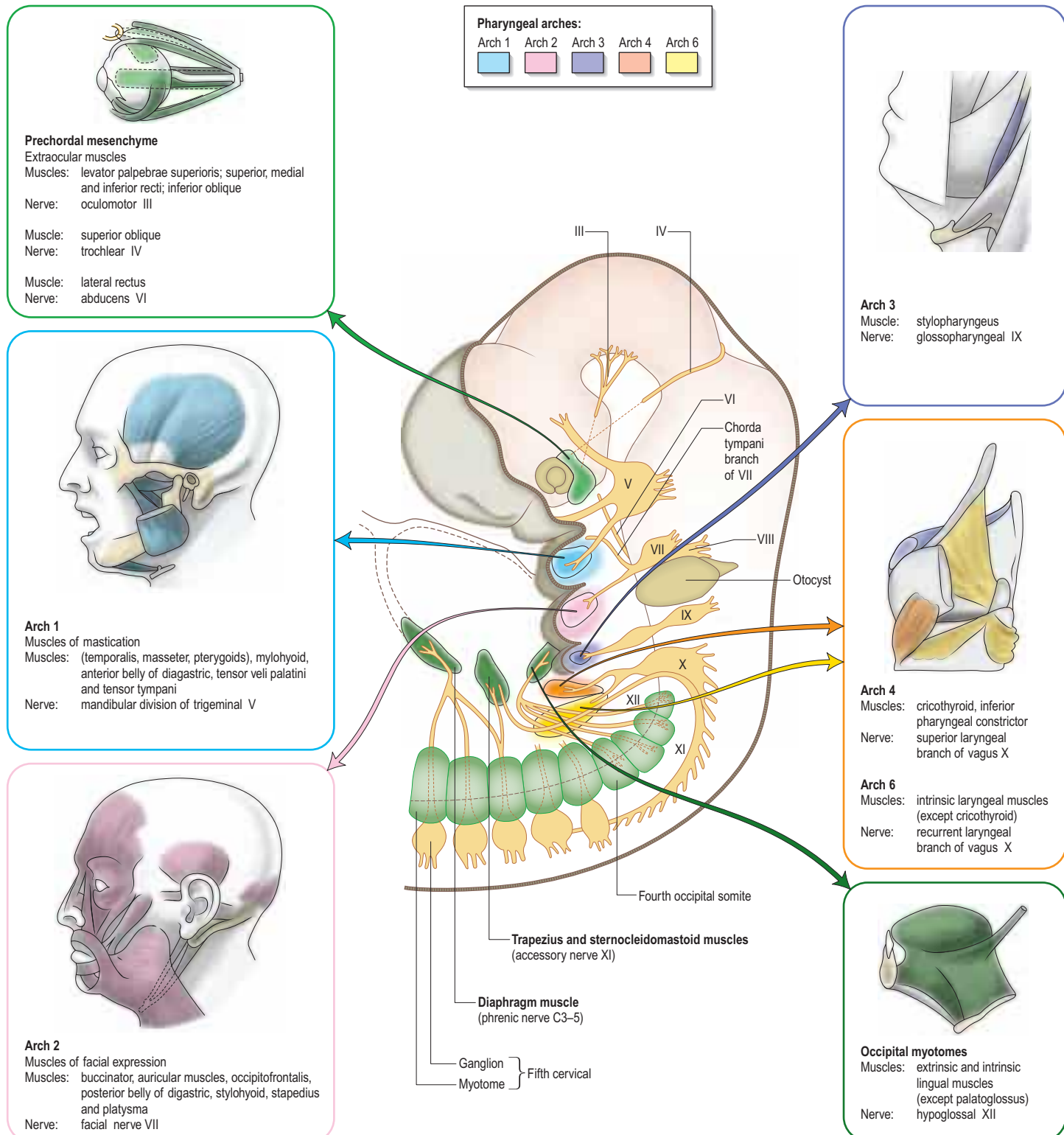


Fig. 36.7 The muscular derivatives of the prechordal mesenchyme, unsegmented paraxial mesenchyme and rostral somites.

constrictors is uncertain. Palatoglossus, a muscle of the palate, is innervated by the vagus via the pharyngeal plexus and may derive from the lower arches.

NERVES OF THE PHARYNGEAL ARCHES

The nerves associated with each arch arise from the adjacent hindbrain (see Fig. 36.7; see also Figs 12.4, 17.18). They are cranial nerves V (trigeminal), VII (facial), IX (glossopharyngeal), X (vagal) and XI (accessory). The motor nerves extend from the basal plate of the hindbrain to innervate the striated muscle of the arches; they are termed special branchial (visceral) efferent nerves because they innervate branchiomeric (pharyngeal) musculature. Sensory nerves extend peripherally and centrally from cranial ganglia that are formed partly from neural crest and partly from cells that delaminate from epipharyngeal placodes; they convey general and special somatic afferent axons. Each arch is innervated by a mixed nerve, but the nerves of the first three arches also have a purely sensory branch that innervates the arch rostral to its 'own' arch; this sensory nerve is called the pretrematic branch because it extends rostral to the cleft or trema between the two arches (see Figs 36.3, 36.7). Hence, the nerves to the mandibular arch include the mandibular division of the trigeminal nerve, which is mixed, and the chorda tympani and greater petrosal nerves, purely sensory pretrematic branches of the facial nerve. The maxillary branch of the trigeminal and the tympanic branch of the glossopharyngeal are also considered to be pretrematic nerves. The ophthalmic branch of the trigeminal nerve, which supplies the frontonasal area, is not an arch nerve; in fishes, its ganglion is separate from the first arch nerve ganglion.

The vagus nerve supplies the fourth and sixth arches (see Fig. 36.7). The recurrent laryngeal branch initially loops under the sixth arch artery on both sides, but the asymmetric changes to the vascular system affect the symmetry of this nerve. On the right, when the sixth aortic arch lateral to the right pulmonary artery degenerates, the recurrent laryngeal nerve loops under the fourth arch-derived subclavian artery; on the left, it loops round the ductus arteriosus, (which is retained as the ligamentum arteriosum in the adult). Further details of cranial nerve development and composition are provided in Chapter 17.

BLOOD VESSELS OF THE PHARYNGEAL ARCHES

The aortic arches initially develop by vasculogenesis, soon after neural crest cells have invaded the early pharyngeal arches. Mesoderm-derived angiogenic mesenchyme forms the endothelial lining of the vessels, and neural crest contributes to the outer layers of the walls (Jiang et al 2000, Yoshida et al 2008). The first aortic arch artery is part of the original vascular circuit that links the aortic sac of the heart to the paired dorsal aortae, blood returning to the heart via the allantoic, vitelline and common cardinal veins (see Fig. 13.1C). As the heart descends relative to the forebrain and other rostral structures, the aortic sac (ventral aortic root) gives rise to paired aortic arches at successively more caudal levels, each of which passes laterally on each side of the pharynx to join the dorsal aortae. The dorsal aortae are not invested by neural crest cells, and nor are the more distal (cranial) parts of the carotid arteries, which form by angiogenesis. The whole complement of aortic arches never coexists – the first aortic arch degenerates through remodelling of the cranial arteries before the sixth is formed.

When the first and second aortic arch arteries begin to regress, the supply to the corresponding pharyngeal arches is derived from a transient ventral pharyngeal artery, which terminates by dividing into mandibular and maxillary branches. The second aortic arch artery also degenerates, remaining only as the stem of the stapedial artery. The early vessel anastomoses with the ventral pharyngeal artery, passing, as it does, through the mesenchyme of the stapedial cartilage, which condenses around it, forming the foramen of the stapes. At this stage, the stapedial artery possesses three branches – mandibular, maxillary and supraorbital – which follow the divisions of the trigeminal nerve (Fig. 36.8). The mandibular and maxillary branches diverge from a common stem.

Further development of the third, fourth and sixth aortic arches (Fig. 36.9) produces the main arteries to the head and the great vessels arising from the heart. The common carotid arises from an elongation of the aortic sac, and the third arch artery becomes the proximal part of the internal carotid artery. The external carotid artery arises as a sprout from the common carotid; it incorporates the stem of the ventral pharyngeal artery, and its maxillary branch communicates with the common trunk of origin of the maxillary and mandibular branches of the stapedial artery and annexes these vessels. The proximal part of the

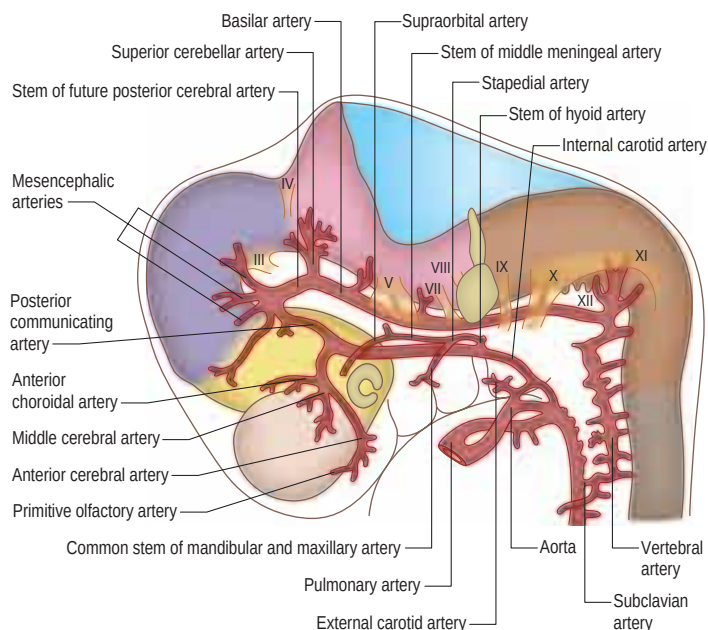


Fig. 36.8 The arterial system of the head at stage 17 (approximately 40 days).

common trunk persists as the root of the middle meningeal artery. More distally, the meningeal artery is derived from the proximal part of the supraorbital artery. The maxillary branch becomes the infraorbital artery and the mandibular branch forms the inferior alveolar artery. When the definitive ophthalmic artery differentiates as a branch from the terminal part of the internal carotid artery, it communicates with the supraorbital branch of the stapedial artery, which distally becomes the lacrimal artery. The latter retains an anastomotic connection with the middle meningeal artery. The dorsal stem of the original second arch artery remains as one or more caroticotympanic branches of the internal carotid artery.

The fourth aortic arch on the right forms the proximal part of the right subclavian artery, whereas the corresponding vessel on the left constitutes the arch of the definitive aorta between the origins of the left common carotid and left subclavian arteries.

The sixth aortic arch is associated with the developing lung buds. Initially, each bud is supplied by a capillary plexus from the aortic sac; as the sixth aortic arch develops, it becomes the channel for blood from the aortic sac to the developing lung buds as well as to the corresponding dorsal aorta, the latter being the main channel. Soon after formation of the sixth aortic arch, the outflow tract of the heart is divided by an influx of neural crest cells, which form the spiral aorticopulmonary septum (Jiang et al 2000); this separates the aortic sac into the pulmonary trunk and ascending aorta. The part of the sixth aortic arch between the pulmonary trunk and the dorsal aorta becomes the main vascular channel on the left, and is defined as the ductus arteriosus when the equivalent vessel on the right side is lost. The right dorsal aorta caudal to the right subclavian artery also degenerates, leaving the left dorsal aorta as the definitive descending aorta (Fig. 36.9B). The ductus arteriosus remains the main channel for blood to pass from the right ventricle to the descending aorta until the lungs and their associated blood vessels expand at birth. After birth, the ductus arteriosus is functionally closed by contraction of the circular muscle of the tunica media (p. 922), leaving a ligamentous remnant, the ligamentum arteriosum (Fig. 36.9C).

PHARYNGEAL ECTODERM AND CLEFTS

Surface ectoderm lines the roof of the embryonic pharynx up to and including Rathke's pouch (see Figs 17.13–17.14). It also completely covers the first arch, including the lateral walls and floor of the pharynx, unlike the more caudal arches, which are lined with pharyngeal endoderm. The external surface ectoderm of the first arch ultimately produces the keratinized stratified squamous epithelium of the epidermis, including hair follicles, sweat and sebaceous glands, and the specialized epithelium of the vermilion of the lip. Within the oral cavity, first arch ectoderm forms the mucous membranes of the internal surface of the lips and cheeks, the palate, the anterior part of the tongue, the epithelial components of the salivary and mucous glands, and the enamel organs

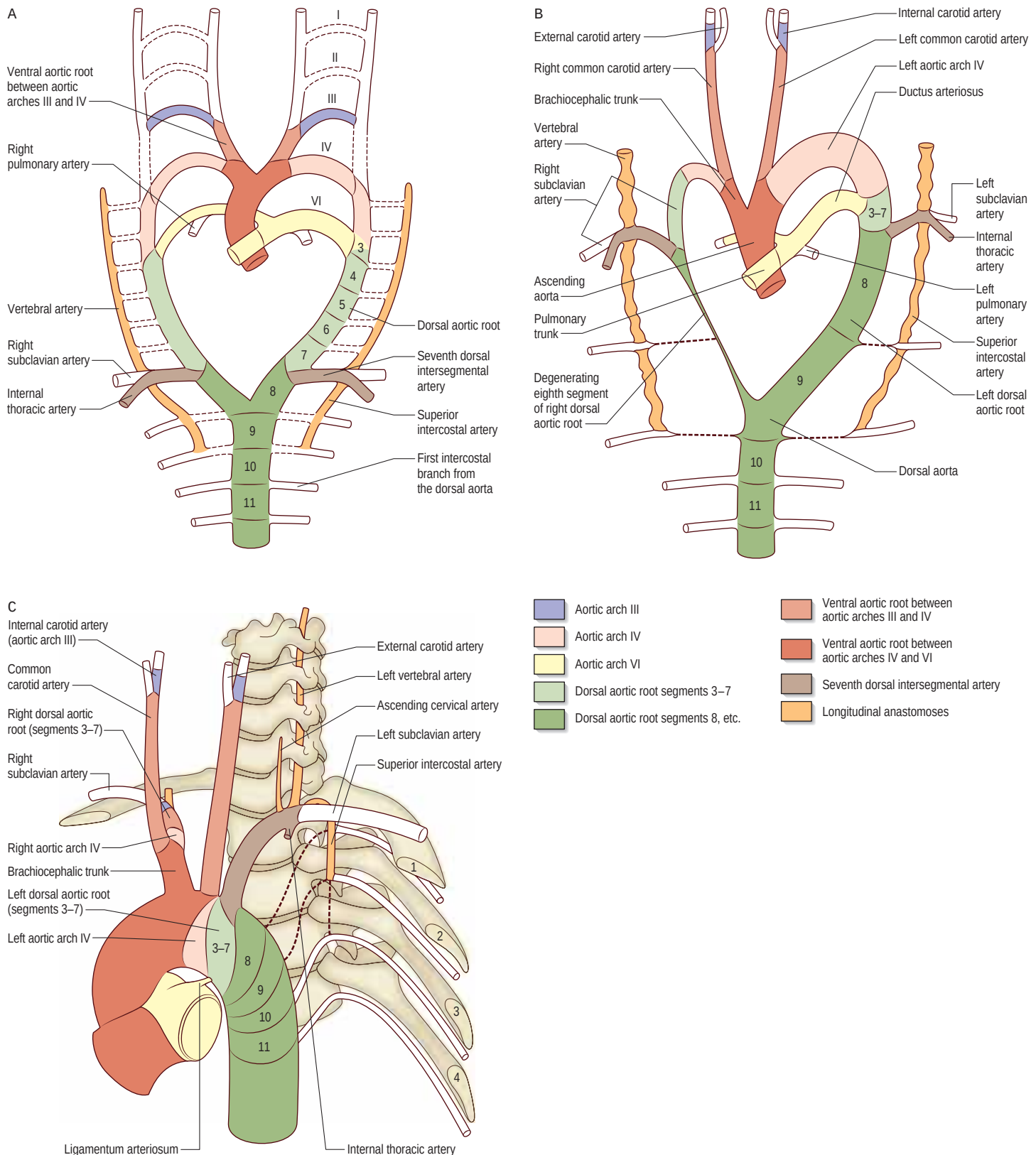


Fig. 36.9 Aortic arch arteries and their derivatives.

of the developing teeth (see Figs 36.14–36.15). The first pharyngeal cleft is obliterated ventrally; its dorsal end deepens to form the external acoustic meatus, including its ceruminous glands and the outer surface of the tympanic membrane (see Fig. 36.4B,C). Three auricular hillocks form on the first and second arches, each side of the first pharyngeal cleft, beginning at stage 15; they form the auricle of the external ear (p. 659).

The external contours of the arches and clefts are modified as the skeletal and muscular elements develop. During the fifth week, the second, third and fourth pharyngeal clefts form the rostral and caudal parts of a retrohyoid depression, the cervical sinus (see Fig. 36.4C). Cranially, the sinus is bounded by the hyoid arch; dorsally, by a ridge produced by ventral extensions from the occipital myotomes and by

mesenchyme that develops into the sternocleidomastoid and trapezius muscles. Caudally, the smaller epiperocardial ridge separates the sinus from the pericardium. Fusion of the hyoid arch with the cardiac elevation covers the cervical sinus, excluding the third, fourth and sixth arches from contributing to the skin of the neck; it also results in platysma muscle, bounded both superficially and deep with superficial fascia, passing along the neck to the anterior thoracic wall.

PHARYNGEAL ENDODERM AND POUCHES

The four pharyngeal pouches appear in sequence craniocaudally during stages 10–13. The rostral pharynx is broad and dorsoventrally

compressed (see Fig. 36.18A). Laterally, the endoderm of the pouches approaches the ectoderm of the pharyngeal clefts to form thin closing membranes (see Fig. 36.4). The approximating ectoderm and endoderm between the first cleft and pouch form the outer and inner surfaces of the tympanic membrane. The ventral end of the first pouch is obliterated, but its dorsal end persists and expands as the head enlarges. This, together with the adjoining lateral part of the pharynx, possibly with a contribution from the dorsal part of the second pharyngeal pouch, constitutes the tubotympanic recess. The recess forms the middle ear cavity, the pharyngotympanic tube and their extensions. Expansion of the middle ear cavity around the ear ossicles occurs late during fetal life, by breakdown of the mesenchyme around the ossicles so that the cavity comes to surround them, except at their attachments to the tympanic membrane and fenestra ovalis (fenestra vestibuli, oval window).

The ectoderm of the pharyngeal clefts and the endoderm of the pouches become increasingly separated by mesenchyme. The blind recesses of the second, third and fourth pouches are prolonged dorsally and ventrally as angular, wing-like diverticula (see Fig. 36.18); the pouch endoderm thickens and evaginates into localized neural crest-derived mesenchymal condensations. Further development of the second, third and fourth pouches is summarized in Figure 36.18 and described in the associated text.

Rarely, failure of normal modification of the pharyngeal clefts and pouches may lead to the formation of fistulae, external or internal sinuses, and cysts (see Commentary 2.2). Lateral cervical cysts are now considered to be lymphoepithelial in origin.

FACE, NASAL CAVITIES AND PALATE, ORAL CAVITY AND TONGUE

FACE

By stage 12, migration of the most rostral neural crest cell population is complete and the sequence of morphogenetic changes that will form the face begins (Fig. 36.10). Neural crest cells have populated the mandibular arch and its maxillary extension, and formed the frontonasal mesenchyme that covers the telencephalon and caudal diencephalon (see Fig. 36.2D). They also contribute to the primordium of the trigeminal nerve, whose first fibres begin to extend at stage 13. The three divisions of the trigeminal nerve will provide sensory innervation to the frontonasal, maxillary and mandibular parts of the face.

Localized mesenchymal proliferation results in the formation of four paired processes: the medial and lateral nasal, maxillary and mandibular processes. The lateral and medial nasal processes surround the nasal placodes, causing them to sink deep to the surface, forming the nasal pits. The two medial nasal processes approach each other to form the nasal septum between them and extend downwards to form the premaxillary component of the upper lip and jaw and the primary palate. Concomitant with formation of the nasal processes, proliferation of the maxillary mesenchyme forms the maxillary processes. These make contact with the medial and lateral nasal processes at stages 16–17 (see Fig. 36.10E). Fusion of the maxillary processes with the medial nasal processes unites the premaxillary and maxillary parts of the upper jaw and lip; failure of this process on one or both sides causes unilateral or bilateral cleft lip (see Fig. 36.17). Superiorly, this fusion closes the cleft at the lower edge of the nasal pits, completing the future nostrils. Fusion of the maxillary with the lateral nasal processes is a much simpler process and rarely gives rise to abnormality. At stage 16, the epithelium of the groove between these fusing processes thickens to form the lacrimal lamina, the primordium of the lacrimal system. At stage 19, it separates from the surface ectoderm, forming the lacrimal cord beneath the surface. The cord becomes canalized to form the nasolacrimal duct in the tenth week (de la Cuadra-Blanco et al 2006). The original site of the lacrimal lamina marks the lateral division between frontonasal and maxillary contributions to the face, the territories whose sensory nerve supply is from the ophthalmic and maxillary divisions of the trigeminal nerve, respectively (Fig. 36.11).

At the start of facial development, the stomodeal opening extends across the whole width of the embryonic head. A wide mouth is maintained until differential growth brings the eyes and the lateral parts of the maxillary–mandibular structures from the sides to the front of the face (stage 18 to stage 23, 7–8 weeks). Although differential growth makes the mouth opening proportionately smaller, progressive fusion of the lateral regions of the maxillary and mandibular processes also makes an important contribution to decreasing the width of the mouth and to forming the cheeks. Differences in mouth width within the human population are mainly due to variation in the extent of maxil-

lomandibular fusion. The period from 7 to 8 weeks is the final period of major morphogenetic change in facial development. In addition to formation of the mouth and cheeks, it is marked by further narrowing of the nasal region and philtrum, and ascent of the ears. By stage 20, the face is recognizably human, although differential growth will continue to bring about changes in proportion and relative position of the features.

The contribution made by the frontonasal process to surface epithelial derivatives extends from the skin of the forehead, over the supraorbital and glabellar regions, including the upper eyelid and conjunctiva, to the external aspect of the nose and philtrum of the upper lip. During later development, the maxillary nerve invades the skin of the philtrum and nasal alae, so that the trigeminal nerve distribution does not completely coincide with the tissue origins (compare Figs 36.10F and 36.11).

NASAL CAVITIES AND PALATE

After formation of the nasal pits, the original olfactory placodal cells lie deep within the pit, adjacent to the underlying telencephalon. They differentiate into olfactory neuroblasts, supporting (sustentacular) cells and stem cells. Neurites begin to extend from the olfactory neuroblasts, reaching the olfactory region of the telencephalon by stage 16 (sixth week). The earliest pioneer neurites cross a mesenchyme-filled gap between the placode and the brain. The remaining placodal cells differentiate into columnar supporting cells, rounded basal cells and, by invagination, the duct-lining and secretory acinar cells of the glands of Bowman. Basal infiltration by lymphocytes is a relatively late event.

Once the nasal cavities have formed, the roof of the oral cavity is regarded as the primitive palate (not to be confused with the primary palate, which forms from the most rostral part of the primitive palate). Each nasal cavity has an arcuate shape, extending upwards over the primary palate to the differentiating olfactory epithelium and then down again towards the oral cavity, from which it is separated by an epithelial choanal membrane (see Fig. 17.13). Towards the end of stage 18, the choanal membranes degenerate to form internal openings, the choanae, lateral to the primary palate; the nasal cavities are now a duct from the external environment to the oral cavity. Within the nasal cavity, the ectodermal lining and subjacent mesenchyme differentiate to form the mucous membrane of the nasal vestibule, conchae and paranasal sinuses, and the olfactory epithelium.

At stage 17, the internal aspect of each maxillary process begins to bulge into the oral cavity as a palatal process; the palatal processes are the primordia of the secondary palate (Fig. 36.12). The tongue is well developed by this stage, and its domed form fills the space between the nascent palatal shelves laterally and the primitive palate above. The mandible is still short, relative to the maxillary region, and does not extend as far as the primary palate. Further growth of Meckel's cartilages brings the mandibular region and tongue forwards, extending to a position below the primary palate. At the same time, the (secondary) palatal processes continue to grow, projecting vertically downwards on either side of the tongue. At stage 23 (56–57 days), the tongue descends and the palatal shelves rotate, bringing the shelves to a horizontal position above the tongue (Fig. 36.12B). The rapidity of the process of shelf elevation is caused, at least in part, by swelling of the palatal mesenchyme due to accumulation and hydration of hyaluronan in the extracellular matrix (Ferguson 1991). This occurs during a period of continuous growth in head height but almost no growth in head width. This latter factor is important because, if palatal shelf elevation is delayed so that it occurs during a period of growth in facial width, the unfused processes may be unable to make contact in the midline, resulting in a cleft palate (see Fig. 36.17). Other factors affecting palatal closure include the growth in length of Meckel's cartilages and the change in position of the maxillae relative to the anterior cranial base; both of these have the effect of lifting the head and upper jaw away from the mandible during weeks 9 and 10, facilitating withdrawal of the tongue from between the palatal shelves and hence creating space for their elevation. Mouth opening, tongue protrusion and hiccup movements have also been noted at this time, and it may be that these movements and their associated pressure changes assist palatal shelf elevation. Generally, palatal shelf elevation occurs slightly later in female embryos than in males.

Midline fusion of the two palatal shelves involves adherence and breakdown (by apoptosis) of the apposed epithelial surfaces. They also fuse with the posterior border of the primary palate, except over a small median area, where a nasopalatine canal maintains connection between the nasal and oral cavities for some time and marks the future position of the incisive fossa. During the period of palatal shelf development, a

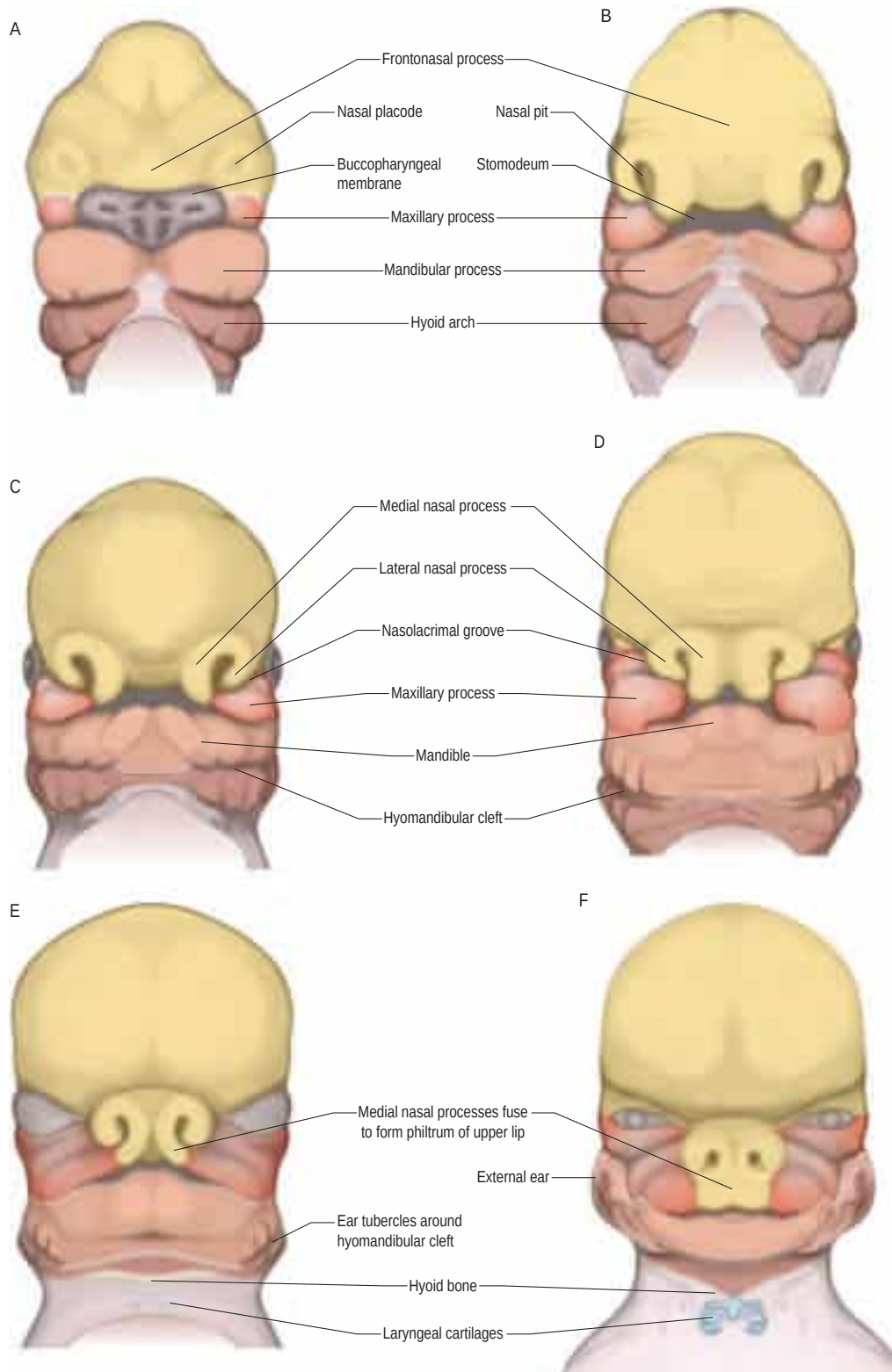


Fig. 36.10 Development of the face during stages 12–18. **A**, 4 weeks (3.5 mm). **B**, 5 weeks (6.5 mm). **C**, 5½ weeks (9 mm). **D**, 6 weeks (12 mm). **E**, 7 weeks (19 mm). **F**, 8 weeks (28 mm).

median downgrowth from the primitive palate forms the definitive nasal septum; it is supported by cartilage. The fusing medial edges of the elevated palatal shelves also fuse with the free edge of the nasal septum, forming separate right and left nasal cavities above the secondary palate. The original positions of the choanae are now covered by the palatal shelves so that the nasal cavities extend from the nostrils to the free (caudal) edge of the palate at the nasopharynx. Bone forms by both intramembranous ossification within the palatal mesenchyme to form the hard palate, except near its caudal border, and by endochondral ossification of the cartilaginous nasal septum. At this time, the free edge of the palatal shelves lies directly below the former position of the attachment of the adenohypophyseal (Rathke's) pouch to the roof of the oral cavity. Caudal to the hard palate, its free edges grow to form the soft palate and uvula, which extend into the oropharynx. Myogenic mesenchyme from the first (tensor palati only) and third pharyngeal arches migrates into the soft palate and around the caudal margins of the pharyngotympanic tube. Separation of the definitive nasal and oral

cavities is now complete. A number of elevations appear on the lateral wall of each nasal cavity: the superior, middle and inferior conchae (Fig. 36.12B,C).

Within the nasal septum, on either side of the septal cartilage, a pair of small diverticula, the vomeronasal organs, form at stage 18 (6 weeks). From 12 to 19 weeks, the vomeronasal organ has an olfactory epithelium-like appearance, and luteinizing hormone-releasing hormone (LHRH)-immunoreactive cells are present in nerve fascicles extending from it to the olfactory bulb (Boehm et al 1994). In many species, the vomeronasal organ has a pheromonal communication function; however, although most adult humans have a vomeronasal organ, its olfactory-type structure and LHRH immunoreactivity are lost during later fetal life and a pheromonal function in adults has not been established (Ch. 33 e-book only).

Bilateral dorsal expansions of the first, second and, possibly, third pharyngeal pouches form the tubotympanic recesses, which will become the tympanic cavities and pharyngotympanic tubes. A number of focal

proliferations of nasopharyngeal endoderm become invaded by lymphoid tissue.

ORAL CAVITY

In the fifth week (stage 12), the oropharyngeal membrane breaks down so that communication is established between the ectoderm-lined stomodeum and the endoderm-lined part of the pharynx, creating a continuous cranial foregut cavity. Formation of the hard and soft palates separates the cranial foregut into the nasal cavity and nasopharynx above and the definitive oral cavity below. The oral cavity is demarcated from the oropharynx posteriorly by the oropharyngeal isthmus, formed by the free edge of the soft palate and uvula above, and the palatoglossal arch laterally.

Tongue

Because the mandibular arch grows more rapidly than the others, it makes the greatest contribution to the pharyngeal floor. By stage 14 (5 weeks), three swellings are apparent: a small median elevation, the tuberculum impar or median tongue bud; and paired lateral lingual swellings distally (Fig. 36.13A). These three swellings fuse to form the anterior two-thirds of the tongue (Fig. 36.13B), and a small median diverticulum, the foramen caecum (thyroid diverticulum), forms immediately caudal to the median tongue bud (Fig. 36.13C; see Fig. 36.18A). A sulcus forms along the ventral and lateral margins of this elevation and deepens, internal to the future dental lamina, to form the linguoingival sulcus (groove) (see Fig. 36.13C). Caudal to the median tongue bud, a small swelling, the copula, forms in the floor of the second arch. The hypobranchial (hypopharyngeal) eminence forms in the floor of the third arch; it grows over the second arch (which does not contribute to the tongue) and fuses with the anterior tongue rudiment along a V-shaped line, the sulcus terminalis, at stage 17 (sixth week). Another transverse groove separates the developing tongue from the caudal (fourth arch) swelling, which forms the epiglottis.

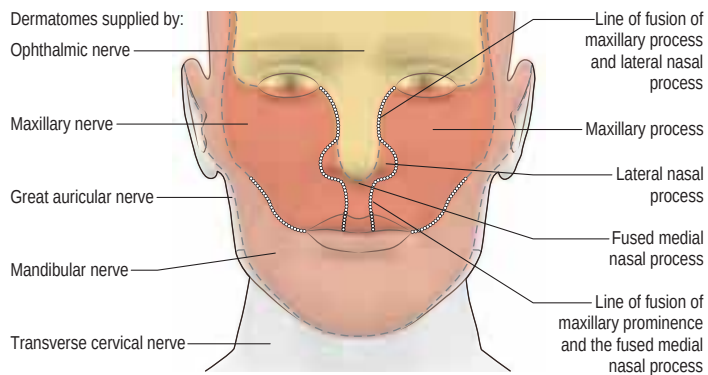


Fig. 36.11 The parts of the adult face derived from the ophthalmic (frontonasal), maxillary and mandibular divisions of the skin of the face, showing the lines of fusion and definitive innervation.

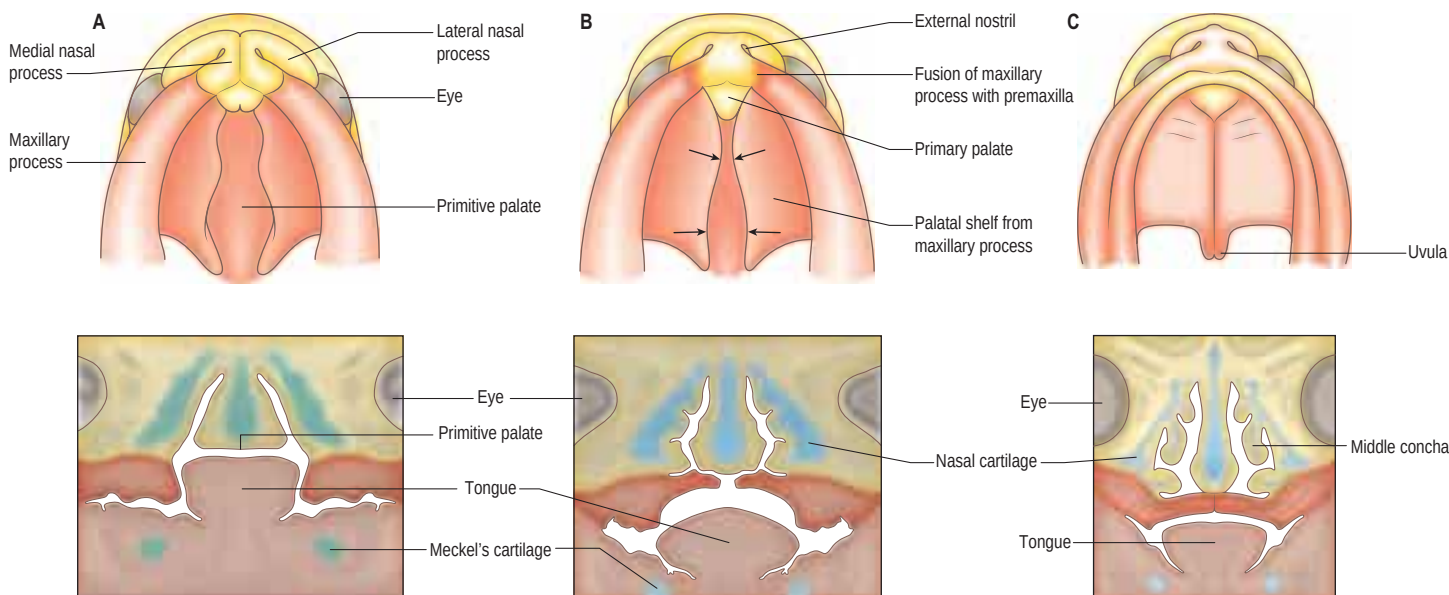


Fig. 36.12 Development of the palate. **A–C**, Developmental progression over weeks 7–8; ventral views (top row) and coronal sections (bottom row).

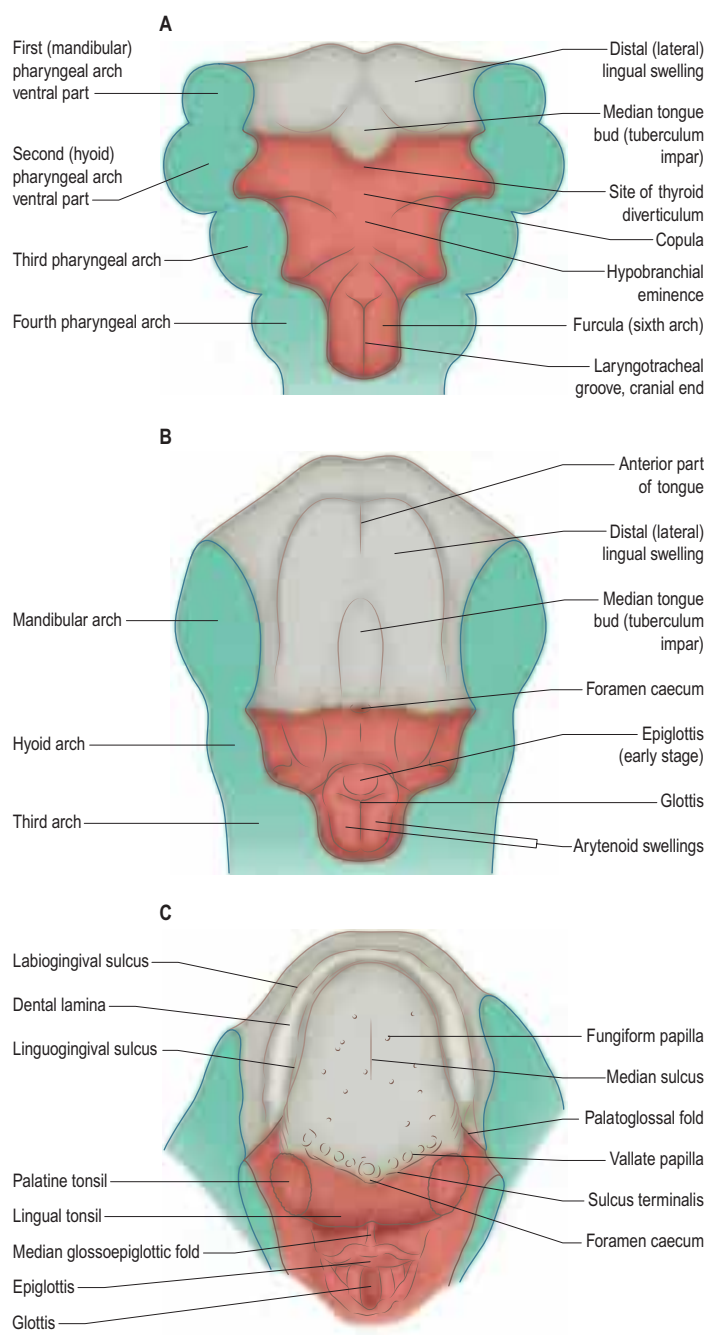


Fig. 36.13 Development of the pharyngeal floor and tongue. **A**, Stage 14, showing median and lateral swellings in the floor of the first arch. **B**, After fusion of the swellings. **C**, The tongue and oropharynx at birth.

In general, the composite character of the mucous membrane of the tongue is reflected in its sensory innervation. The anterior, oral, part is innervated by the lingual branch of the mandibular nerve, and by the chorda tympani of the facial nerve. The posterior, pharyngeal, part of the tongue is innervated by the glossopharyngeal nerve, which is the nerve of the third arch; this nerve also invades the tissue immediately distal to the sulcus limitans, including innervation of the vallate papillae. The vallate papillae appear within the mucous membrane at around 10 weeks and increase in number until the end of the second trimester. The root of the tongue, near the epiglottis, is innervated by the vagus. The motor innervation is from the hypoglossal nerves, which extend around and under the pharynx together with myogenic cells migrating from the occipital myotomes during stages 14 and 15.

Tongue development is essentially complete by stage 23. At birth, the tongue is short and broad, and its entire surface lies within the oral cavity (see Fig. 14.7). The posterior third of the tongue descends as the hyoid bone and larynx descend during the first postnatal year, and by the fourth or fifth year, it forms part of the anterior wall of the oropharynx.

Salivary glands

The salivary glands arise bilaterally as the result of epithelial-mesenchymal interactions between the ectodermal epithelial lining of the oral cavity and the subjacent neural crest-derived mesenchyme. The parotid gland can be recognized at stage 15 (8 mm) as an elongated furrow running dorsally from the angle of the mouth between the mandibular and maxillary prominences. The groove, which is converted into a tube, loses its connection with the epithelium of the oral cavity, except at its ventral end, and grows dorsally into the soft tissue of the cheek. The tube persists as the parotid duct. After fusion of the lateral parts of the maxillary and mandibular prominences, the parotid duct opens on the inside of the cheek at some distance from the angle of the mouth. As the gland develops, its branches interweave with the branches of the facial nerve. In the neonate, the parotid gland is rounded and lies between masseter and the ear. During infancy and early childhood, the growing gland extends to cover the parotid duct. Its secretions remain entirely serous.

The submandibular and sublingual glands form as solid diverticula that undergo branching morphogenesis, the whole tree-like structure later acquiring a lumen. The blind ends of the branches form acini, whose cells differentiate to form serous cells initially, and mucus-secreting cells postnatally. The submandibular gland is identifiable at stage 18 as an epithelial diverticulum into the mesenchyme from the floor of the caudal part of the linguogingival sulcus. It increases rapidly in size and branches several times. At first, the connection of the submandibular gland with the floor of the mouth lies at the side of the tongue, but as the edges of the linguogingival sulcus come together,

from behind forwards, they enclose the elongating duct. The orifice of the submandibular duct is thus located progressively more distally until it is below the tip of the tongue, close to the median plane. The sublingual gland arises at stage 20 as a number of small epithelial thickenings within and lateral to the linguogingival sulcus. Each thickening canalizes separately, so multiple sublingual ducts open on the summit of the sublingual fold, while others join the submandibular duct.

The minor salivary glands, which are distributed throughout the wall of the oral cavity, except for the gingiva and hard palate, form in a similar manner to the major glands but undergo very little branching and remain within the submucosa.

Lymphoid tissues

Tonsils form at several sites around the oro- and nasopharynx, where focal proliferations of endoderm become invaded by lymphoid tissue. The endodermal epithelial lining grows into the surrounding mesenchyme as a number of solid buds, which are excavated by degeneration and shedding of their central cells, forming tonsillar fossae and crypts. Lymphoid cells accumulate around the crypts at about the fifth month and become grouped as lymphoid follicles; T- and B-cell regions can be identified. The palatine tonsils develop from the ventral parts of the second pharyngeal pouches (see Fig. 36.4C). A slit-like intratonsillar cleft extends into the upper part of the tonsil and is possibly a remnant of the second pharyngeal pouch. Lymphoid tissue similar to that of the palatine tonsils is found in the first pouch (tubal tonsils), the surface of the posterior part of the tongue (lingual tonsils), and in the dorsal pharyngeal wall (adenoid or pharyngeal tonsil).

Teeth and gums

Demarcation of the lips begins after fusion of the facial primordia at stage 18, during the period of secondary palatal development. Two parallel epithelial thickenings, an outer labiogingival lamina and an inner dental lamina, form around the interior border of the mouth in both upper and lower jaws (see Figs 36.13C, 36.15). The linguogingival sulcus is formed as an indentation between the dental lamina and the tongue. At stage 22, each labiogingival lamina indents the underlying mesenchyme to form a shallow groove that deepens to form the labiogingival sulcus between the lips and gums. The lining of the oral cavity between the vestibular epithelium of the upper and lower jaws forms the inner aspect of the cheeks. The dental lamina gives rise to the epithelial component of the teeth.

Tooth formation involves a sequence of epithelial-mesenchymal interactions along the dental lamina (Fig. 36.14). These interactions are controlled by a large number of genetic factors, and the extracellular

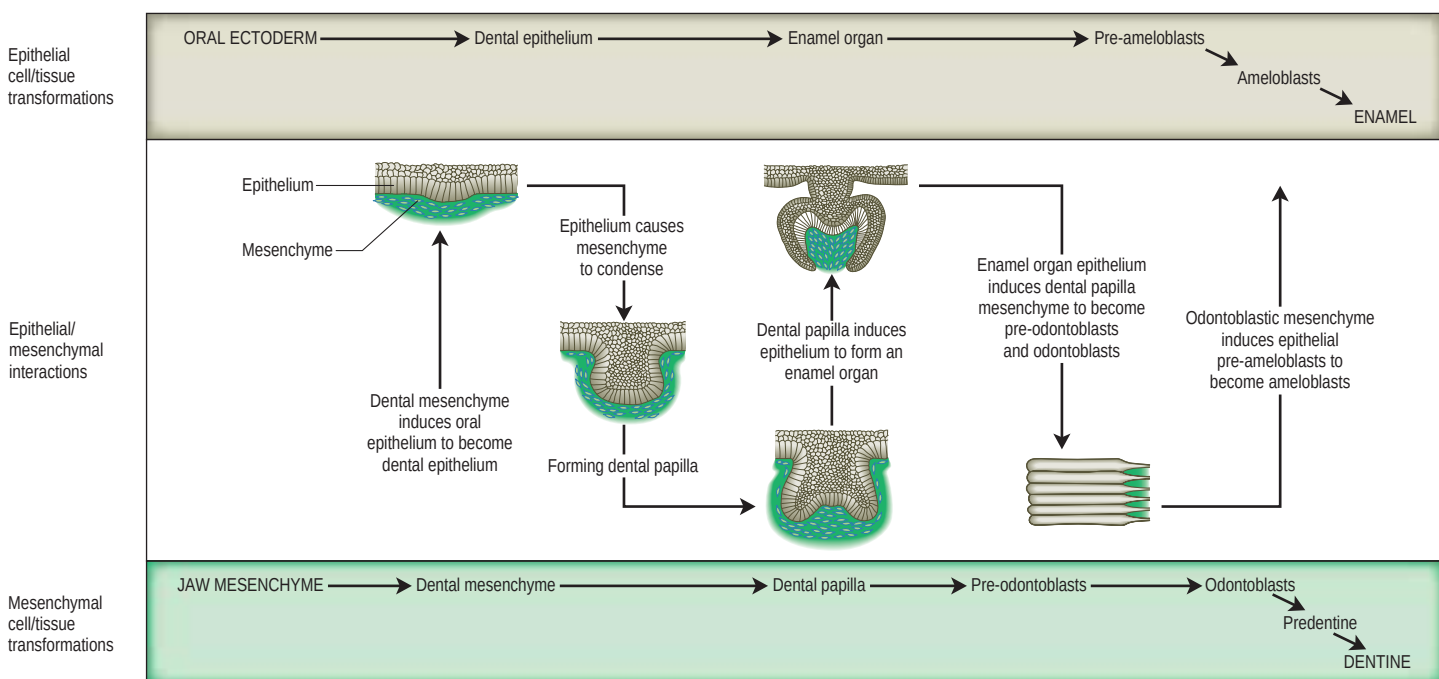


Fig. 36.14 A summary of the epithelial-mesenchymal interactions during tooth development.

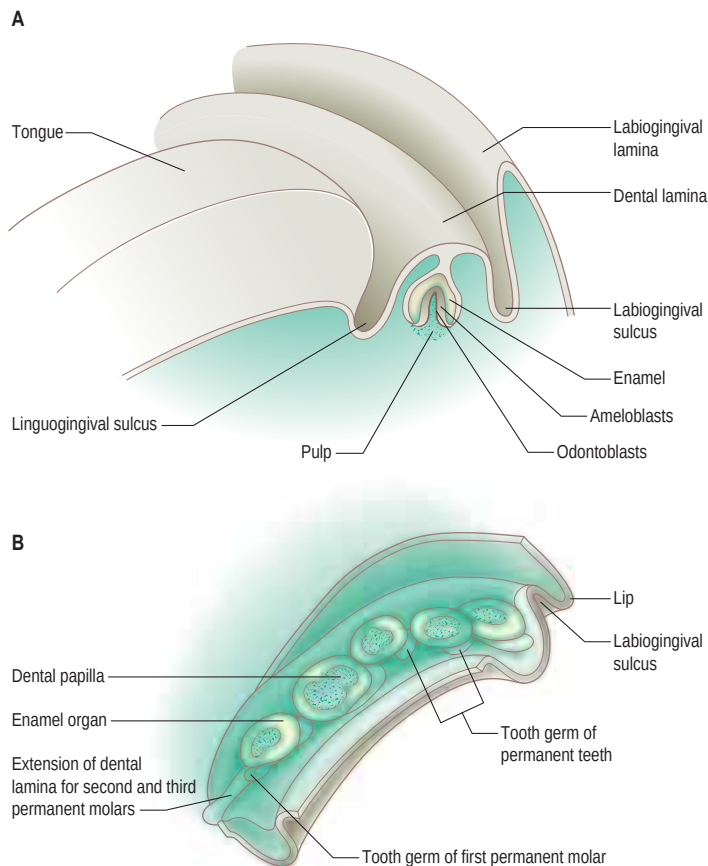


Fig. 36.15 The lip, gums and teeth after 24 weeks. **A**, Oral view. **B**, The right half of the lower jaw viewed from below; the midline is upper right.

matrix (particularly of the basement membranes) is of major importance (Thesleff 2000, Fukumoto and Yamada 2005). In 25 mm embryos, 10 localized thickenings of the upper and lower dental laminae initiate formation of the dental epithelial components of the tooth primordia (see Fig. 36.14). They expand into dental sacs surrounded by vascular mesenchyme. The epithelium proliferates and indents to form an enamel organ, which, by 10 weeks, forms a cap over a mesenchymal condensation, the dental papilla; collectively, this unit constitutes a tooth bud or germ. The enamel organ becomes bell-shaped and differentiates into two layers, the internal and external enamel epithelia; these are separated by a glycosaminoglycan-rich extracellular matrix, the stellate reticulum. The cells of the internal epithelium differentiate into ameloblasts, and the underlying layer of mesenchymal cells differentiates into odontoblasts. The odontoblasts produce and secrete dentine, which influences the ameloblasts to form enamel; this is laid down in successive layers and, after mineralization, is the hardest tissue in the body. Enamel and dentine production from the bell-shaped tooth germ both begin at 24 weeks (Fig. 36.15). The dental papilla mesenchyme below the odontoblast layer forms the pulp, which becomes vascularized and innervated. Experimental studies in mice (which have only incisors and molars) have revealed that regional differences in signaling pathways within the maxillary and mandibular mesenchymes govern the type of tooth produced (Tummers and Thesleff 2009). Fibronectin is present in the basement membrane of the internal enamel epithelium during the bell phase. It assists attachment of the pre-odontoblasts and is essential for their differentiation; it becomes progressively more abundant during the later fetal period. Collagen I is the major supportive extracellular scaffold in the maturing dental papilla and stellate reticulum.

At birth, each quadrant of the jaws has five deciduous tooth germs and the germs of the first five permanent teeth. Permanent teeth develop in accessional positions from the lingual aspects of the deciduous tooth germs and, for the 12 permanent molar teeth, from posterior extensions of the dental laminae on each side of both jaws. Mineralization begins in both deciduous and permanent teeth before birth. The deciduous teeth have well-developed crowns by full term, and eruption through the gingiva (gum) usually begins with the lower incisors between 3 and 6 months after birth. After eruption, the gingiva remains attached to the neck of each tooth by the upper part of the periodontal ligament, whose major part attaches the tooth root to the bone of the tooth socket.



Fig. 36.16 Pierre Robin sequence. (Courtesy of Mr T Goodacre and Oxford Radcliffe Hospitals NHS Trust.)

The oral cavity after birth

In the neonate, the oral cavity is only potential when the mouth is closed, but three spaces are formed during suckling. The median space between the tongue and hard palate, which is occupied by the nipple, bifurcates posteriorly to produce channels on each side of the soft palate and epiglottis. The larynx is elevated so that its opening is directed into the nasopharynx, i.e. above the level of the channels, and milk passes on either side of it to the pharynx and oesophagus. This anatomical arrangement enables the infant to breathe while suckling (see Figs 34.11, 34.12A). The hard palate is only slightly arched and its mucous membrane is corrugated by five or six irregular transverse folds (rugae), which assist with gripping the nipple. Two lateral spaces, the lateral arcuate cavities, are formed between the tongue and cheeks; the upper and lower gums situated within these spaces do not touch during suckling. Each cheek is supported by a mass of subcutaneous fat, the suctorial pad, which lies between buccinator and masseter.

ANOMALIES OF FACIAL DEVELOPMENT

Malformations involving the neural crest

Congenital malformations of the face may be initiated early, mainly related to abnormal migration, proliferation and/or apoptosis of neural crest cells, or later, during the morphogenetic phase of facial development. Craniofrontonasal syndrome is characterized by a broad forehead and hypertelorism, premature fusion of one or both coronal sutures, a central nasal groove, and sometimes cleft lip and palate; there are also extracranial features. It is caused by deletions of the gene *EFNB1*, which is known to play essential roles in boundary formation between tissues, including those between neural crest and adjacent non-crest cells (Twigg et al 2004). Mandibulofacial dysostosis (Treacher Collins syndrome) is characterized by incomplete orbits, reduced jaws and auditory ossicles (hence conductive deafness), abnormal external ears and fistulae or cysts, suggesting involvement of the first and second arch components of the cranial neural crest. It is caused by haploinsufficiency of the gene *TCOF1*. Hemifacial microsomia (Goldenhar's syndrome) affects only one side of the face and comprises a small mandible (sometimes with absence of the temporomandibular joint), malformed or absent ears with accessory ear tags, and facial clefts. Experimental evidence in animals suggests that it is due to haemorrhage of the stapedia artery, which may have a genetic basis. Deficient growth of the mandible (micrognathia) is usually associated with other defects. In Pierre Robin sequence (Fig. 36.16), the small size of the mandible obstructs descent of the tongue, resulting in a U-shaped cleft palate; a conotruncal septum defect is present in one-quarter of cases, suggesting possible involvement of the cardiac neural crest. Further details and references for facial malformations may be found in Morriss-Kay and Wilkie (2005) and Hennekam et al (2010).

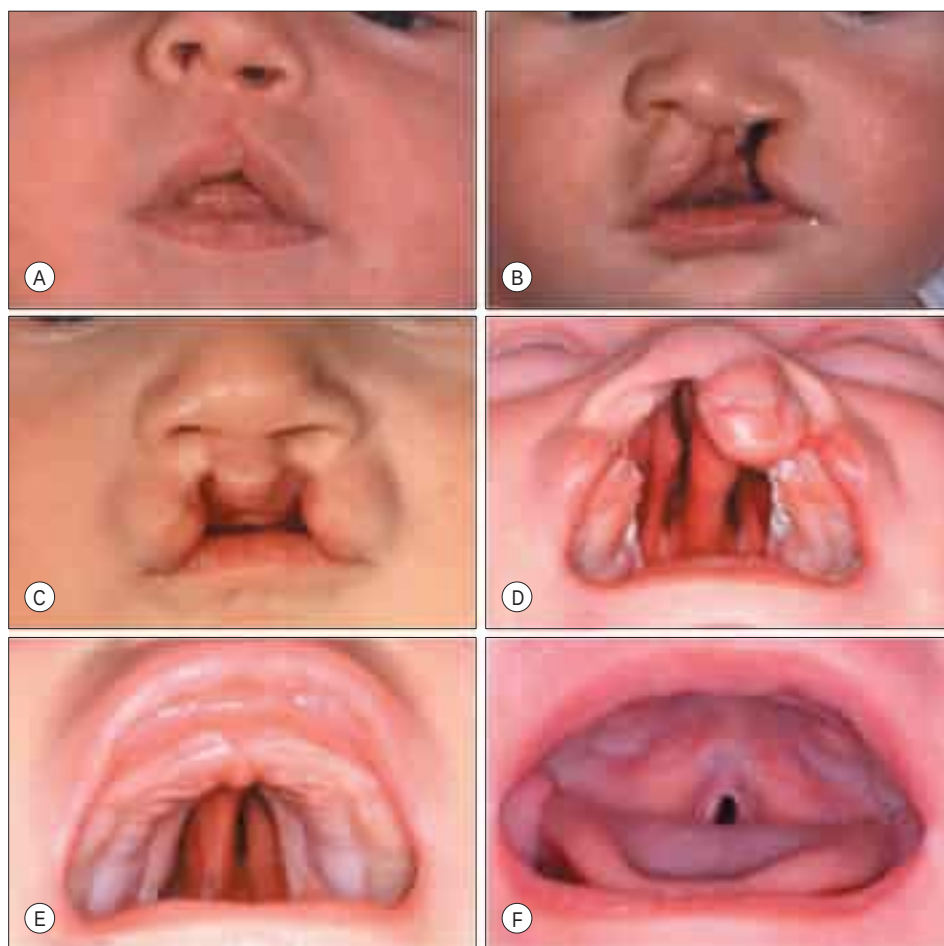


Fig. 36.17 Cleft lip and cleft palate. **A**, Incomplete unilateral cleft lip. **B**, Complete unilateral cleft lip. **C**, Incomplete bilateral cleft lip. **D**, Complete bilateral cleft lip and cleft palate (the nasal septum and both inferior nasal conchae can be seen). **E**, Cleft palate (the nasal septum and left inferior nasal concha can be seen). **F**, Cleft of the soft palate. (Courtesy of Mr T Goodacre and Oxford Radcliffe Hospitals NHS Trust.)

Clefts of the face and palate

Cleft lip occurs when one or both medial nasal processes fail to fuse with the corresponding maxillary process(es). The severity of clefting varies from a small notch in the upper lip to a double cleft extending into both nostrils (Fig. 36.17). In a unilateral cleft lip, the nasal septum deviates to the non-cleft side. This is because the muscles on the medial side of the cleft (transverse muscles of the nose and orbicularis oris) are not inserted into either the muscles of the contralateral side or the tissues around the nasal septum (Fig. 36.17B). The three functional groups of superficial facial muscles – nasolabial (transverse nasalis, levator labii superioris and levator labii superioris alaeque nasi), labial (orbicularis oris) and labiomental (depressor anguli oris) – are all displaced inferiorly. A further consequence is the underdevelopment of the incisor-bearing part of the maxilla. These abnormalities, in turn, influence the mucocutaneous tissues, which results in the displacement of the skin of the nostril to the upper part of the lip, retraction of the labial skin and abnormalities of the soft tissues on either side of the mucocutaneous junction.

A narrow or incomplete cleft of the hard palate is likely to be due to a growth defect in the palatal shelves and/or a delay in shelf elevation, and/or to failure of fusion of the apposed palatal shelves. Broader clefts (Fig. 36.17D,E), in which the tongue lies between the still-vertical shelves, are caused by a greater developmental delay that may be associated with abnormality of other structures in the face and elsewhere in the body. In the mildest forms of cleft palate, only the soft palate, and sometimes just the uvula, is cleft (Fig. 36.17F). A submucous cleft, which may be detected as a median V-shaped notch of the soft palate, can cause speech difficulties. Very rarely, muscle hypoplasia, particularly of the musculus uvulae, causes palatopharyngeal incompetence.

Cleft lip with or without cleft palate (CL/P) is multigenic or multifactorial in origin, i.e. it has both genetic and environmental components (Gorlin et al 2001). CL/P has a frequency of 0.5–3.6/1000 live births, with a 2 : 1 male to female ratio. Isolated cleft palate is genetically unrelated to CL/P, and has a frequency of 0.4/1000; it is more common in females. Clefts of the face are rare (1/50,000–1/175,000 births); they are mainly caused by pressure from amniotic bands rather than from dysmorphogenesis, although a morphogenetic component cannot be ruled out when the cleft is along a line of fusion, e.g. maxillary–mandibular. Median cleft lip (true hare lip) is also rare, since this is not a line of morphogenetic fusion. Median grooving of the nose occurs

with hypertelorism in various frontonasal dysplasias, suggesting a developmentally early broadening of the median region of the face.

FORMATION OF THE NECK

By the time the facial swellings have merged to form a recognizably human face (stage 20, 7 weeks), Meckel's cartilage has grown sufficiently to project outwards as a potential chin but the face is still orientated towards, and close to, the cardiac prominence (see Fig. 36.5). Formation of the posterior part of the neck is, by this time, well advanced, with cartilaginous cervical vertebrae, muscles, nerves and blood vessels. The anterior neck forms gradually, as relatively rapid growth of the anterior structures, including the trachea, oesophagus, common carotid arteries and jugular veins, takes place. As the extending neck lifts the head to a forward-facing orientation, the angle between the spinal cord and brainstem increases. Other contributory changes include growth of the first pharyngeal arch at a more rapid rate than the others, so that Meckel's cartilage and the developing membrane bone of the mandible project further forwards, while the hyoid bone primordium remains small, defining the future angle between the anterior neck and lower jaw. Myoblasts derived from paraxial mesenchyme invade the second pharyngeal arch as it forms and by this stage are present in the superficial connective tissue beneath the anterior neck skin; these spread out from the underside of the chin to the upper thorax and differentiate to form the platysma (this is the main human remnant of the panniculus carnosus, the muscle that is present throughout the skin of many mammals).

The interface between cranial nerve-innervated pharyngeal arch muscles and spinal nerve-innervated trunk muscles occurs at the lower neck and the upper parts of the pectoral girdle. Both the proximal and distal connections of the trapezius muscle, at the nuchal line of the occipital bone and the attachment to the spine of the scapula, are formed from post-otic neural crest cells (Matsuoka et al 2005). The neural crest generates areas of endochondral ossification in the cervical vertebral column and scapula associated with trapezius attachment. This observation has promoted the conception of a 'muscle scaffold model' of development to replace the older 'ossification model', to explain the types of ossification seen associated with the accessory nerve-innervated trapezius and sternocleidomastoid muscles. It is suggested that defects in the fate choices of neural crest cells could explain

a number of syndromes that have pharyngeal, laryngeal, occipital, cervical and shoulder dysmorphologies and co-occurrence of swallowing problems; related syndromes include Klippel–Feil, Sprengel's deformity and cleidocranial dysplasia (Matsuoka et al 2005) (see p. 794).

GLANDS OF THE NECK

Further development of the pharyngeal floor and the second to fourth pharyngeal pouches gives rise to glands that are ultimately disposed in the neck and anterior mediastinum, namely: the paired thyroid and parathyroid glands and the thymus.

Thyroid gland

The thyroid gland is a midline derivative of the pharynx. It is first identifiable in embryos of approximately 20 somites as a median thickening of endoderm lying in the floor of the pharynx between the first and second pharyngeal pouches and immediately dorsal to the aortic sac. The foramen caecum is the site of a median diverticulum that appears early in the fifth week in the furrow immediately caudal to the median tongue bud (see Fig. 36.13B, C; Fig. 36.18). It extends caudally as the thyroglossal duct, passing ventral to the primordium of the hyoid bone. The tip of the duct bifurcates and the tissue mass subsequently divides to form the isthmus and lateral lobes of the thyroid gland. The endoderm-derived epithelial tissue is invested by vagal neural crest mesenchyme, which gives rise to the connective tissue capsule and interlobular septa, and perifollicular mesenchyme, which carries the neurovascular and lymphatic supply to the gland. The gland has three stages of development. Stage 1 (10–18 weeks) is characterized by active folliculogenesis and gradual accumulation of colloid; the thyroid tissue

is able to concentrate iodine and synthesize thyroxine. Stage 2 (19–29 weeks) is quiescent, with little change in the epithelium/colloid ratio and follicular size. Stage 3 (after 29 weeks) is characterized by a gradual increase in the epithelium/colloid ratio and a decrease in follicular size (Bocian-Sobkowska et al 1997). The parafollicular or C cells of the thyroid gland are derived from the ultimobranchial body, a small diverticulum of the fourth pharyngeal pouch. The thyroid gland is relatively large in the neonate (see Fig. 14.6A), where it has a long, narrow isthmus connecting lobes that do not yet contact the upper part of the trachea. It usually attains half the adult size by 2 years of age.

Failure of downgrowth of all or part of the thyroid gland from the pharyngeal floor results in ectopic thyroid tissue within the tongue (lingual thyroid), lying between the foramen caecum and the epiglottis. Occasionally, parts of the midline thyroglossal duct persist and may occur in lingual, suprahyoid, retrohyoid or infrahyoid positions; they may form aberrant masses of thyroid tissue, cysts, fistulae or sinuses, usually in the midline. Nodules of glandular tissue may also be found laterally, posterior to sternocleidomastoid.

Parathyroid glands

The parathyroid glands develop from interactions between the third and fourth pharyngeal pouch endoderm and local cranial (vagal) neural crest mesenchyme. The third pharyngeal pouch has dorsal and ventral sites of proliferation (Fig. 36.18B). The epithelium on the dorsal aspect of the pouch and in the region of its duct-like connection with the cavity of the pharynx becomes differentiated as the primordium of the inferior parathyroid glands (parathyroid III). Although the connection between the pouches and the pharynx is soon lost, the connection between the dorsal parathyroids and the ventral thymic rudiments of the third pharyngeal pouch persists for some time and both move caudally. The

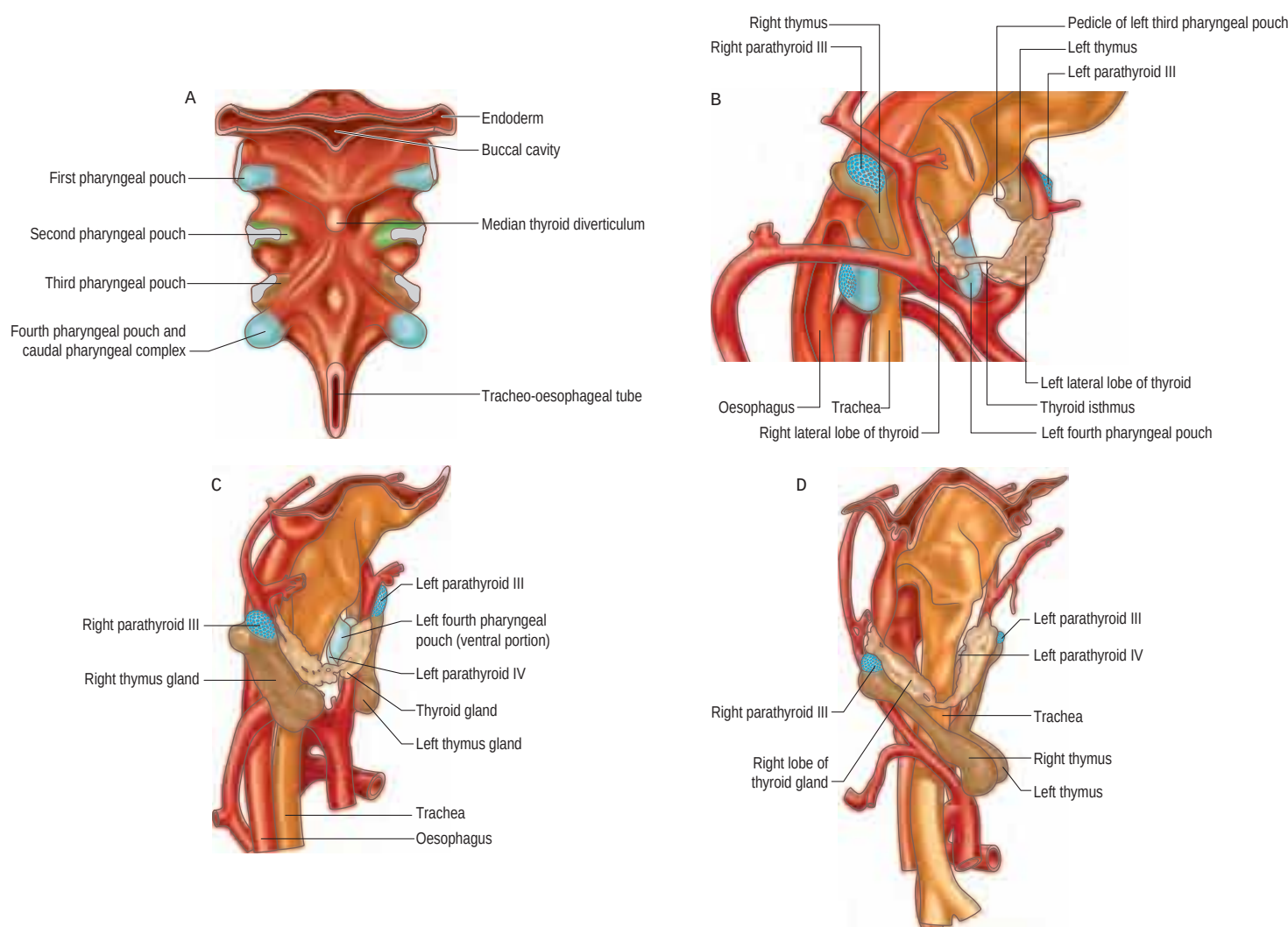


Fig. 36.18 **A**, The ventral aspect of the endoderm of the pharynx, showing the pharyngeal pouches. The areas of contact of the pharyngeal endoderm with the surface ectoderm are shown as flattened surfaces. Note that the colours of the pharyngeal pouches and the median thyroid diverticulum are retained in **B**, **C** and **D**. **B**, Ventral and dorsal diverticuli of the third and fourth pharyngeal pouches and midline thyroid gland at 6 weeks. **C**, The thymus, thyroid and parathyroid glands at 7 weeks. **D**, The thymus, thyroid and parathyroid glands at 7½ weeks. (Redrawn with permission from Hamilton WJ, Boyd JD, Mossman HW 1962 Human Embryology: Prenatal Development of Form and Function. Cambridge: W Heffer & Sons.)

superior parathyroid glands (parathyroid IV) develop from the dorsal recess of the fourth pharyngeal pouches. They make contact with, and appear to be anchored by, the lateral lobes of the thyroid gland and thus remain cranial to the descended parathyroid III glands. Vagal neural crest mesenchyme provides the connective tissue elements, and invading angiogenic mesenchyme gives rise to fenestrated capillaries and lymphatics. In the neonate, the parathyroid glands are as variable in size and position as they are in the adult. They double in size between birth and puberty. Parathyroid hormone is produced from the twelfth week of development.

Thymus

The thymus gland is formed from the ventral part of the third pharyngeal pouch on each side (see Fig. 36.18). It cannot be recognized prior to the differentiation of the inferior parathyroid glands at stage 16 but, thereafter, it is represented by two elongated diverticula that soon become solid cellular masses and grow caudally into the surrounding vagal neural crest mesenchyme. Ventral to the aortic sac, the two thymic rudiments meet but do not fuse, and they are subsequently united by connective tissue only. The thymus gland becomes located in the anterior mediastinum of the thorax once the neck is fully developed and the heart has descended. The connection with the third pouch is soon lost, but the stalk may persist for some time as a solid, cellular cord.

As the thymus proliferates and descends, the local neural crest-derived mesenchyme controls the pattern and development of the gland. Defective development of the vagal neural crest, which affects the heart and peripheral neural ganglia, also results in thymic deficiencies, as seen in DiGeorge syndrome and Pierre Robin sequence. Neural crest mesenchyme forms connective tissue septa, which produce the lobulated architecture of the gland. Angiogenic mesenchyme, including lymphoid stem cells, invades this local mesenchyme, and by 10 weeks, over 95% of the cells in the gland belong to the T-cell lineage, with a few erythroblasts and B lymphocytes. Hassall's corpuscles are also present. By 12 weeks, the mesenchymal septa, blood vessels and nerves have reached the newly differentiating medulla, which allows the entry of macrophage lineage precursors. Macrophages and interdigitating cells are first seen at 14 weeks. Granulopoiesis occurs in the perivascular spaces. By 17 weeks, the thymus appears fully differentiated, and after this time, it produces the main type of thymocyte that is present throughout life (designated Td⁺).

Caudal pharyngeal complex

The most caudal endodermal evaginations of the pharynx are the fourth pharyngeal pouch and the ultimobranchial body, a small diverticulum medial to the main pouch. Collectively, these diverticula are termed the caudal pharyngeal complex (see Fig. 36.18), and they are connected to the pharynx via the pharyngobranchial duct. They are surrounded by vagal neural crest and by the tissues of the developing thyroid gland. The cells of the ultimobranchial body become incorporated into the lateral thyroid lobes, and give rise to the calcitonin-secreting 'C' or parafollicular cells of the thyroid gland.

PHARYNX, LARYNX, OESOPHAGUS AND TRACHEA

The larynx forms at the cranial end of the laryngotracheal groove, caudal to the epiglottis (see Figs 36.5–36.6, 36.13). The thyroid cartilage develops from the ventral ends of the cartilages of the fourth, or fourth and sixth, pharyngeal arches. The cartilage appears as two lateral plates, each chondrified from two centres and united in the mid-ventral line by a fibrous membrane, within which an additional centre of chondrification develops. The cricoid cartilage arises from two cartilaginous centres, which soon unite ventrally, gradually extend, and ultimately fuse on the dorsal surface of the tube as the cricoid lamina. It has the appearance of a modified tracheal cartilage. Paired arytenoid swellings appear within the ventral tissue of the sixth arches from stage 14, one on each side of the cranial end of the laryngotracheal groove. As they enlarge, they approximate to each other and to the caudal part of the hypobranchial eminence, where the epiglottis develops. The opening into the larynx, at first a simple slit, is converted into a T-shaped cleft by the enlargement of the arytenoid swellings. The vertical limb of the T lies between the two swellings, and its horizontal limb lies between them and the epiglottis. The arytenoid swellings differentiate into the arytenoid and corniculate cartilages (Fig. 36.6), and the ridges that join them to the epiglottis become the definitive aryepiglottic folds, within which the cuneiform cartilages differentiate from the

epiglottis. Two linear ridges in the ventral wall of the pharynx extend ventrally from the sixth arches to the median laryngotracheal groove, from which the lower part of the larynx, the trachea, bronchi and lungs develop (Ch. 52).

The laryngotracheal groove forms at stage 12 as a ventral endodermal outgrowth from the floor of the caudal part of the pharynx into the mesenchyme surrounding the sinus venosus and inflow tract of the heart (see Figs 52.1C, 60.3). This initiates the separation of the respiratory and alimentary tubes. The groove remains at a constant level during the embryonic period, and the trachea lengthens distally as its bifurcation point descends. The splanchnopleuric mesenchyme that surrounds the developing trachea and oesophagus induces the different histogenic pathways of the two tubes. By stage 17, the mesenchyme around the trachea is beginning to condense at regular intervals and undergo chondrogenesis to form C-shaped cartilages, whereas that around the oesophagus has a wide submucosal zone and muscular coats. The respiratory diverticulum becomes surrounded by angiogenic mesenchyme that connects to the developing sixth aortic arch artery (see above).

For details of early development of the trachea, see page 926; for early development of the oesophagus, see page 1048.

In the neonate, the pharynx is one-third of the relative length seen in the adult. The nasopharynx is a narrow tube that curves gradually to join the oropharynx without any sharp junctional demarcation. An oblique angle is formed at this junction by 5 years of age, and in the adult, the nasopharynx and oropharynx join at almost a right angle. The oesophagus in the newborn begins and ends one to two vertebrae higher than in the adult, extending from between the fourth and sixth cervical vertebrae to the level of the ninth thoracic vertebra. The narrowest constriction is at its junction with the pharynx, where the inferior pharyngeal constrictor functions to constrict the lumen; in this region, it may be traumatized with instruments or catheters.

The neonatal hyoid bone and larynx are relatively high in the neck, and the trachea is relatively small in relation to the larynx (see Fig. 14.6B); both descend during infancy. The walls of the trachea are relatively thick and the tracheal cartilages are relatively closer together than in the adult. The trachea commences at the upper border of the sixth cervical vertebra, a relationship that is conserved with growth, and it bifurcates at the level of the third or fourth thoracic vertebra.

BLOOD VESSELS IN THE NECK

With elongation of the neck and the appearance of the major conducting vessels from the aortic arch arteries, longitudinal anastomoses in the cervical region link intersegmental arteries and their branches, and direct blood flow to the developing brain in the vertebral arteries in parallel with the internal carotid arteries (see Figs 36.8–36.9).

The primary blood vessels of the head and neck consist of a close-meshed capillary plexus drained on each side by the precardinal vein, which is, at first, continuous cranially with a transitory primordial hindbrain channel that lies on the neural tube medial to the cranial nerve roots (see Figs 13.1–13.2). This is soon replaced by the primary head vein, which runs caudally from the medial side of the trigeminal ganglion, lateral to the facial and vestibulocochlear nerves and the otocyst, then medial to the vagus nerve, to become continuous with the precardinal vein. A lateral anastomosis subsequently brings it lateral to the vagus nerve.

The ventral pharyngeal vein drains the mandibular and hyoid arches into the common cardinal vein (see Fig. 17.36A). As the neck elongates, its termination is transferred to the cranial part of the precardinal vein, which later becomes the internal jugular vein (see Fig. 17.36B). The ventral pharyngeal vein receives tributaries from the face and tongue, and becomes the linguofacial vein. As the face develops, the primitive maxillary vein extends its drainage into the territories of supply of the ophthalmic and mandibular divisions of the trigeminal nerve, including the pterygoid and temporal muscles, and it anastomoses with the linguofacial vein over the lower jaw. This anastomosis becomes the facial vein; it receives blood from the retromandibular vein from the temporal region, and drains through the linguofacial vein into the internal jugular. The stem of the linguofacial vein is now the lower part of the facial vein, whilst the dwindling connection of the facial with the primitive maxillary becomes the deep facial vein. The external jugular vein develops from a tributary of the cephalic vein from the tissues of the neck; it anastomoses secondarily with the anterior facial vein. At this stage, the cephalic vein forms a venous ring around the clavicle, by which it is connected with the caudal part of the precardinal vein. The deep segment of the venous ring forms the subclavian vein and receives the definitive external jugular vein. The superficial segment of the venous ring dwindles, but may persist in adult life. The deep

aspects of the maxillomandibular facial prominences, the retrogingival oral cavity, the pharyngeal walls and their lymphoid and endocrine derivatives, and the cervicothoracic oesophagus thus all have drainage channels that connect with the precardinal complex. Laryngeal and tracheobronchial veins also drain to the precardinal complex, whilst the capillary plexuses, developed in the (splanchnopleuric) walls of the fine terminal respiratory passages and alveoli, converge on pulmonary veins of increasing calibre, finally making secondary connections with the left atrium of the heart, and may be grouped with the vitelline systems.

SKULL

The skull has two major functional and anatomical components: the neurocranium and the viscerocranium. The neurocranium is composed of the calvaria and basicranium; it surrounds and protects the brain and the special sense organs of olfaction, vision, hearing and balance. The viscerocranium, which includes the squamous part of the temporal bone, forms the skeleton of the face, palate and pharynx, and mediates the functions of feeding, breathing and facial expression; it also protects the tongue and forms the middle ear and the bony external acoustic meatus. The most rostral neural crest cell population makes a major contribution to the skull, forming the whole viscerocranium and the rostral part of the neurocranium (Fig. 36.19). The boundary between neural crest and cranial mesoderm lies between the frontal and parietal bones (coronal suture) of the calvaria; the skull base is formed by neural crest rostral to the tip of the notochord, and is sclerotome-derived (i.e. from paraxial mesenchyme) in the notochordal region (McBratney-Owen et al 2008). The tip of the notochord lies immediately caudal to the hypophyseal fossa (Fig. 36.20A). Broadly speaking, the bones of the skull base are formed by endochondral ossification (chondrocranium), whereas those of the calvaria and face ossify directly from mesenchymal condensations, i.e. by intramembranous ossification. Several bones are of compound structure with respect to their tissue origins and/or type of ossification: the occipital, temporal and sphenoid bones, and the mandible. The following account is based largely on descriptive studies on human embryos and fetuses (O'Rahilly and Müller 1987, 2001, 2007), with some details of cell lineage based on investigations in the mouse.

CHONDROCRANIUM

The first cranial skeletal structures to differentiate are the cartilages of the skull base, sensory capsules, viscerocranium and occiput (see Figs 36.5, 36.20B). Development of the pharyngeal arch-related cartilages of the viscerocranium and ear ossicles has been described in the section on the pharynx. Before any mesenchyme condenses to initiate the formation of cranial cartilage, the major cranial nerves and blood vessels are already in place; hence, the foramina of the skull are specified before

the bones are formed. The adenohypophyseal (Rathke's) pouch is still connected to the roof of the oral cavity as the skull base cartilage forms (dashed lines in Fig. 36.20A; see also Fig. 17.14), but already lies adjacent to the diencephalon-derived neurohypophysis to form the downwardly projecting pituitary gland (hypophysis) at the base of the caudal diencephalon.

The notochordal sheath is rich in sulphated glycosaminoglycans, which play a role in inducing condensation and chondrification of the occipital sclerotome-derived mesenchyme around it.

The structures referred to in the following account are illustrated in Figure 36.20B, which shows the cartilaginous skull base of a 40 mm (9-week) embryo. The central regions of all four occipital sclerotomes contribute to the parachordal cartilage, which surrounds the notochord and extends as a flat plate on either side of it by the end of the seventh week, forming the basioccipital component of the occipital bone (see Fig. 44.13). Laterally, the exoccipital components (derived from sclerotomes 3 and 4) chondrify soon afterwards; they extend around the hindbrain to form the occipital arch, which is developmentally equivalent to the neural arch components of vertebrae. Roots of the hypoglossal nerve run between the parachordal and exoccipital cartilages, so that when the exoccipital and parachordal components fuse, they leave foramina for these nerve roots (see Fig. 44.12). The supraoccipital part of the occipital cartilage extends dorsally from the exoccipital cartilage to complete the foramen magnum. After formation of the exoccipital cartilages, differentiation also extends further rostrally in the medial part of the skull base, with formation of the hypophyseal polar cartilages on either side of the hypophyseal stalk; they unite in the median plane to form the primordium of the postsphenoid, cradling the hypophysis and retaining a perforation for the hypophyseal stalk until the third month. This part of the basisphenoid cartilage will form the sella turcica with its hypophyseal fossa. The presphenoid cartilages, rostral to it, form the jugum of the sphenoid body. This is the last part of the medial part of the skull base to differentiate as cartilage, bridging the gap between the postsphenoid and the cartilaginous nasal capsule.

At stage 17, mesenchyme begins to condense and later to chondrify around the nasal pits, forming the outer part of the nasal capsule and the nasal septum; the roof of each nasal capsule is completed slightly later, when cartilage differentiates around the olfactory nerve bundles to form the cribriform plates of the ethmoid bone. During the third month, cartilaginous conchae form. The whole nasal capsule is well developed by the end of the third month, and consists of a common median septal part, sometimes initially termed the interorbitonasal septum or mesethmoid, and the outer ectethmoid. The conchae ossify during the fifth month; the superior and middle conchae form part of the ethmoid bone, and the inferior pair become separate elements. Each lateral part of the nasal capsule becomes ossified as the orbital plate and ethmoidal labyrinth, whose spaces ('cells') communicate with the nasal cavity and become filled with air after birth (see p. 566). Part of the capsule remains cartilaginous as the septal and alar cartilages of the nose, and part is replaced by the intramembranous vomer and nasal bones.

Orbital cartilages surround the eye but do not undergo ossification in their entirety. The most rostral of them becomes continuous with the presphenoid cartilage by differentiation of a cartilaginous bridge that forms the caudal boundary of the optic foramen, enclosing the optic nerve; this later ossifies to form the lesser wing of the sphenoid bone (orbitosphenoid). The greater wing of the sphenoid (alisphenoid) has both intramembranous and endochondral components; the endochondral part initially differentiates as a cartilage surrounding the mandibular branch of the trigeminal nerve, forming the foramen ovale. This condensation extends medially to join the rostral edge of the hypophyseal (polar) cartilage on each side. It also extends rostrally to surround the maxillary branch of the trigeminal nerve, forming the foramen rotundum. Lastly, it extends laterally to join the intramembranous part of the bone, which replaces the caudal part of the orbital cartilage. The greater and lesser wings of the sphenoid are separated by the oculomotor, trochlear and abducens nerves and by the first (ophthalmic) division of the trigeminal nerve.

The otic capsule differentiates from a mesenchymal condensation around the otocyst, after its morphogenesis to form the cochlea and semicircular canals. Differentiation of cartilage begins laterally, at the same time as the hypophyseal cartilages are first detectable; it is complete by stage 20. Chondrogenesis around the point of exit of the vestibulocochlear nerve creates the internal acoustic meatus. Chondrogenesis of mesenchyme around the carotid arteries joins each hypophyseal cartilage to the otic capsule, forming the carotid canals. A gap occupied by the jugular vein and the glossopharyngeal, vagus and accessory nerves remains between each otic capsule and the parachordal cartilage; this is the jugular foramen. Chondrogenesis of the mesenchyme

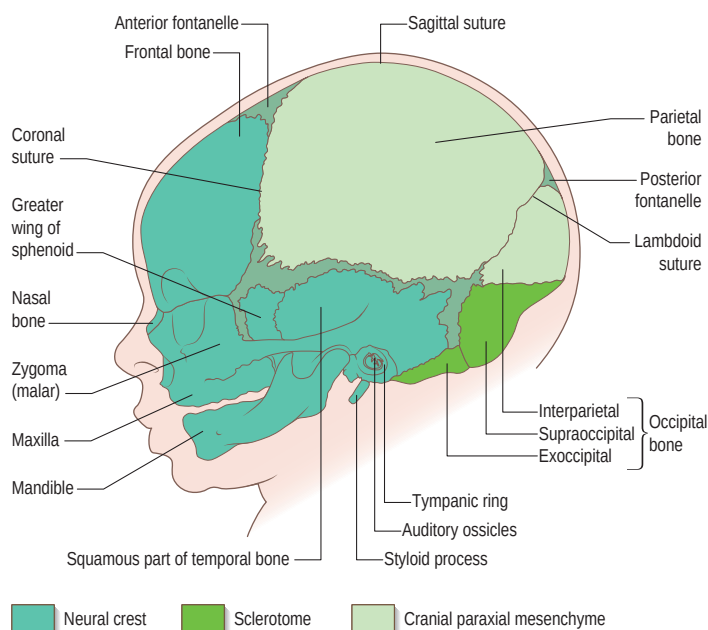


Fig. 36.19 The newborn skull, showing the tissue origins of the bones (based on combined mouse and human data). The darker green represents the sites of fontanelles.

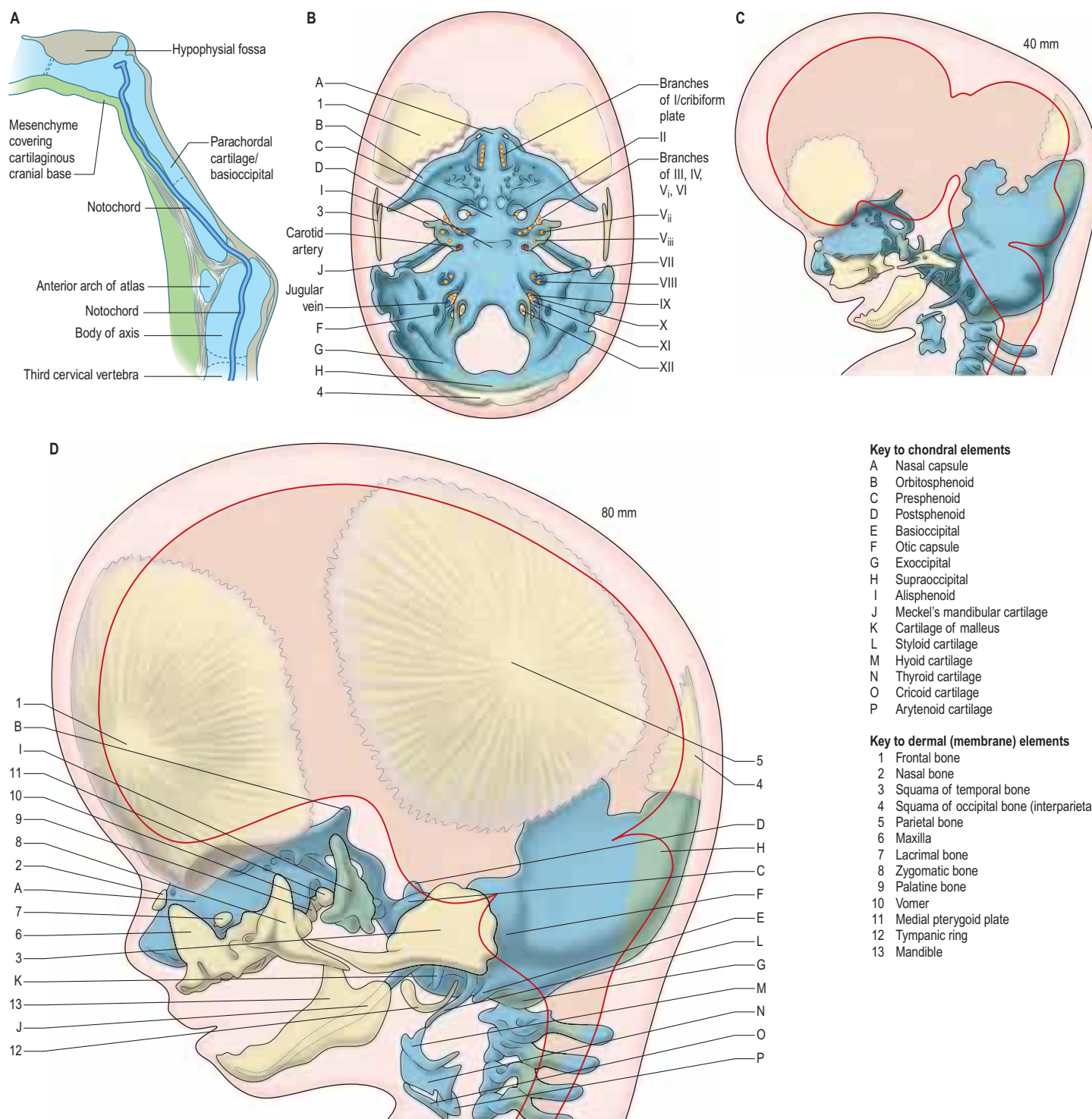


Fig. 36.20 Representative stages in the development of the cranium. In all the diagrams, the chondrocranium and cartilaginous stages of vertebrae are shown in blue, except where ossification is occurring; here, the colour is green. The membrane bones are shown in cream. Cranial nerves are indicated by the appropriate Roman numeral. **A**, A sagittal section through the cranial end of the developing axial skeleton in an early human embryo of approximately 10 mm (5 weeks), showing the extent of the notochord. **B**, The superior aspect of the cranium of a human embryo at 40 mm (9 weeks). The remnant of the hypophysial stalk is indicated by a black dot in the hypophysial fossa. **C**, The lateral aspect of **B**. **D**, The lateral aspect of the cranium of a human embryo at 80 mm.

between the separate cartilages completes formation of the cartilaginous skull base and sensory capsules, which together form a continuous framework around the pre-existing blood vessels and cranial nerves by 9 weeks (40 mm) (see Fig. 36.20B,C).

The process of endochondral ossification in the skull is essentially the same as that of the long bones, except that each ossification centre is equivalent to a primary centre, and follows a specific programme of growth and patterning. Growth takes place within the cartilage between adjacent ossification centres; these joints are termed synchondroses. The first ossification centres to appear are those of the lesser and greater wings of the sphenoid, at 8 and 9 weeks, respectively. A single basisphenoid ossification centre appears at 11 weeks; three presphenoid and four postsphenoid ossification centres appear at 16 weeks. Bone replaces

most of the cartilage until only the major synchondroses remain; these enable the endochondral skull base and sensory capsules to continue growing until they reach their final size at puberty or earlier. At birth, unossified chondrocranium persists in the alae, lateral nasal cartilage and septum of the nose, the sphenoethmoidal junction, the spheno-occipital and spheno-petrous junctions, the apex of the petrous temporal bone (foramen lacerum), and between ossification centres of the sphenoid and occipital bones. As indicated above, parts of the chondrocranium do not ossify, but are replaced by adjacent intramembranous bone, e.g. the greater wing of the sphenoid (alisphenoid). In general, endochondral ossification centres form later than intramembranous ossification sites, which first appear during the seventh and eighth weeks.

CALVARIA

The bones of the calvaria (roof and lateral walls of the neurocranium) are formed entirely by intramembranous ossification. They are also described as dermal bones, since they are considered to be the evolutionary descendants of dermal plates formed as a protective cover for the brain in fishes. In humans, as in all mammals, the major part of the calvaria is formed by paired frontal and parietal bones and the unpaired interparietal (membranous part of the occipital bone). The squamous part of the temporal bones and the alisphenoids contribute to the lateral walls. Lineage data from mouse studies indicate that the frontal and squamous temporal bones are of neural crest origin and the parietals are of mesodermal origin; the interparietal is mixed (Jiang et al 2002). The coronal suture thus forms at the neural crest–mesoderm interface, as does the sagittal suture, due to a small tongue of neural crest tissue lying between the two developing bones. These tissue interfaces are significant for initiating the molecular signalling system that governs growth of the calvaria. Growth at the borders of adjacent bone, i.e. in the sutures, is the major mechanism of calvarial growth. A sequence of events that maintains a balance between cell proliferation and osteogenic differentiation is mediated by an intercellular signalling system that includes the transcription factor TWIST and ligand–receptor interactions between fibroblast growth factors (FGFs) and their receptors (FGFRs). Mutations in the genes encoding these proteins cause premature fusion of the cranial sutures (craniosynostosis), which results in deficient growth in the plane perpendicular to the suture (Fig. 36.21; Morriss-Kay and Wilkie 2005, Johnson and Wilkie 2011). In addition to the expansion-related growth that takes place in the sutures, appositional growth, in which bone is laid down on, and resorbed from, the bone surfaces, plays an important role in remodelling the calvarial bones to maintain a degree of curvature that matches the curved surface of the growing brain. Appositional growth can also provide some compensatory expansion of the skull in craniosynostosis (see Fig. 36.21).

The frontal and parietal bones are initiated as basolateral mesenchymal primordia, which extend upwards between the dermal connective tissue and the mesenchymal dura mater. They do not, as previously thought, differentiate within the mesenchyme of an 'ectomeninx' that surrounds the brain. As the frontal and then the parietal primordia extend upwards, the differentiating osteoblasts secrete osteoid, which then undergoes mineralization. Only the mineralized parts of the bones are detected in alizarin-stained specimens and X-rays. Clear frontal bone primordia are detectable in the superciliary arch region of alizarin-stained human embryos by the eighth week. As they become mineralized above this level, the parietal bone primordia can also be seen. The

two bones appear to be separated by a wide gap at the coronal suture; in fact, this gap only shows the separation of the mineralized parts of the bone – the unmineralized edge of the parietal bone actually overlaps the caudal edge of the frontal bone. After the upgrowth phase, mineralization takes place centrifugally from the central points of the frontal and parietal bones, which, by this time, form convex plates over the curvature of the underlying brain (see Fig. 36.20D).

During vaginal birth, the overlap of the frontal and parietal bones enables them to slide over each other. During the neonatal period, when the baby cries and intracranial pressure rises, the overlapping bones slide apart, increasing the fronto-occipital diameter of the skull. Growth at the coronal suture provides the major increase in the fronto-occipital plane of the calvaria. Premature synostosis of the coronal sutures restricts growth in this plane, causing brachycephaly; unilateral coronal synostosis leads to plagiocephaly, in which one side of the skull fails to grow in the fronto-occipital plane. Uni- and bilateral coronal synostosis comprises 20–25% of all forms of craniosynostosis.

The metopic and sagittal sutures are formed when the upwardly extending frontal and parietal bones (respectively) reach the vertex of the skull and abut in the median plane. Growth in these sutures increases the breadth of the skull. Premature fusion results in the formation of a narrow, elongated skull. Sagittal synostosis is the most common form of craniosynostosis (40–55%).

The interparietal bone (supranuchal squamous portion of the occipital bone) forms from two ossification centres that appear in the eighth week. These are considered to be homologous with the paired post-parietal bones of reptiles. The fact that they unite to form a single bone in mammals may be due to the interpolation of a small area of neural crest that migrates from the hindbrain after neural tube closure (Jiang et al 2002). Two further ossification centres develop laterally at 12 weeks. There is, at first, a wide gap occupied by cartilage between the interparietal and parietal bones, which disappears as the membrane bones grow towards each other, forming the lambdoid suture when they abut. The lambdoid suture contributes growth to the caudal border of the parietal bones and to the upper part of the occipital bone. Synostosis of this suture is relatively rare (fewer than 5% of craniosynostosis cases) and has a less severe effect on overall skull growth than coronal and median fusions. Between the interparietal bone and the foramen magnum, the calvaria is completed by endochondral ossification of the supraoccipital component of the occipital bone.

In addition to the sutures that are formed where two membrane bones abut, fontanelles are formed where three or four bones meet. In the median plane, these are the anterior fontanelle, at the junction of the metopic and sagittal sutures, and the posterior fontanelle, at the junction of the sagittal and lambdoid sutures. The anterolateral

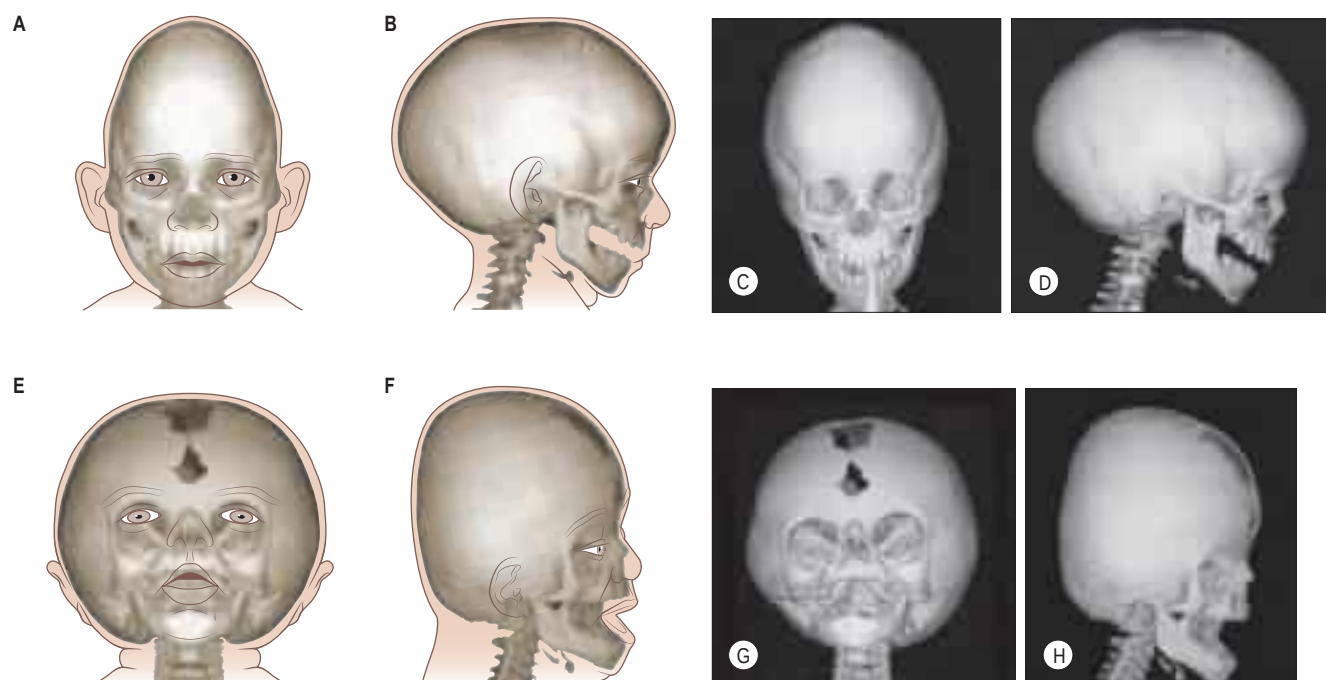


Fig. 36.21 Craniosynostosis. **A–B**, Line drawings of frontal and lateral views of sagittal synostosis. **C–D**, Three-dimensional computed tomography (3D-CT) scans. Premature fusion of the sagittal suture has restricted growth in breadth of the skull but there is compensatory growth in the fronto-occipital plane. **E–F**, Line drawings of frontal and lateral views of bicoronal synostosis. **G–H**, 3D-CT scans. Premature fusion of both coronal sutures has restricted growth in the fronto-occipital plane but there is compensatory growth in breadth. The pairs of CT scans are not from the same patients as those indicated in the line drawings; the endotracheal tube has been digitally removed from D. (CT scans courtesy of P Anslow.)

fontanelle lies between the frontal, parietal, greater wing of the sphenoid and squamous temporal bones; its site after closure is called the pterion. The posterolateral fontanelle lies between the parietal, petrous temporal, exoccipital and basioccipital bones; after closure, its site is called the asterion. The size of the fontanelles at birth, and the timing of their closure, are highly variable. Delayed growth of the skull bones causes ossification defects, including cranium bifidum and parietal foramina, for which several genetic defects have been identified. Cleidocranial dysplasia is a defect of ossification affecting the intramembranous part of the clavicle as well as the calvaria; it is caused by haploinsufficiency of the bone master gene *RUNX2*.

MEMBRANE BONES OF THE FACE AND VISCEROCRANIUM

The face and viscerocranium are formed from neural crest-derived membrane bones. The facial skeleton includes, from forehead to chin: the frontal bones, the orbital bones (frontal, lacrimal and zygomatic), the nasal bones and the vomer, the maxilla and the mandible. The maxilla and mandible form from first arch mesenchyme, as do the medial pterygoid plates of the sphenoid bone, the palatine and the tympanic bones.

During migration, the trigeminal neural crest cells divide into a frontonasal population that migrates superior to the eye and comes to surround the telencephalon and part of the diencephalon, a mandibular population that migrates into the first pharyngeal arch, and a maxillary population that forms the maxillary swelling (see Fig. 36.2). The frontal, lacrimal, nasal bones, the vomer and the premaxillary (incisor tooth-bearing) part of the maxilla are derived from the frontonasal mesenchyme; the maxilla and zygoma are derived from the maxillary mesenchyme; and the mandible and tympanic bone are derived from the mandibular mesenchyme. The mandible is the first membrane bone to begin ossification; its single ossification centre appears in the seventh week. The maxilla and premaxilla have primary ossification centres by 7 weeks, and three further ossification centres (orbitonasal, nasopalatine and zygomatic) form in the maxillary mesenchyme at 8 weeks; these components fuse to form a single bone, in contrast to some mammals, in which the incisor-bearing premaxillary part remains separate. In the neonatal skull, the suture between the primary (premaxilla-derived) and secondary parts of the palate is still patent. By 8 weeks, ossification centres for most of the facial and viscerocranial bones are present, except for the tympanic ring, for which four ossification centres appear at 12 weeks. They fuse to form the sickle-shaped bone that supports the tympanic membrane (see Fig. 36.6). There is as much anteroposterior as vertical growth in the first 5 years of life. After the age of 5 years, there is more vertical growth. During the first 5 years of life, there is a steady increase in maxillary volume; maxillary growth rate accelerates during the time that the permanent dentition is developing and erupting (Langford et al 2003).

The mandible is a developmentally complex bone. Although the body of the bone is formed by intramembranous ossification, the coronoid region and condyle are formed by endochondral ossification in cartilage that develops after formation of the membrane bone. These secondary ossification centres form at 10–14 weeks. Distal cartilage forms two further secondary ossification centres, the mental ossicles, at 7 months. This complex system enables the mandible to grow at both proximal and distal ends, analogous to a long bone.

Growth of the face occurs in the sutures between the membrane bones in a similar manner to that of the calvarial sutures but, during

the fetal period, the amount of growth is proportionately less than that of the calvaria, reflecting the dominant influence of brain growth in the latter.

ANTENATAL ULTRASOUND IMAGING OF THE HEAD AND NECK

The fetal head appears oval on transthalamic and transventricular planes, symmetrical and broader posteriorly (see Fig. 14.4). Biparietal diameter and head circumference are routinely measured. The size of the cisterna magna can be estimated to note hypoplasia of the posterior fossa of the skull. Increased thickness of the nuchal tissues at the back of the neck, measured from the outer edge of the occipital bone to the outer skin edge, is part of screening for Down's syndrome (Nafziger and Vilensky 2014). Nuchal translucency is increased in a number of trisomies and in fetuses with congenital heart defects (Chen 2010). The causes of this are complex and include an increase in hyaluronic acid in the extracellular matrix in the nuchal skin folds, aberrant jugular lymphatic flow and disturbed venous–lymphatic differentiation leading to local oedema (de Mooij et al 2010).

Prenatal diagnosis of asymmetrical craniodystosis and of craniofacial syndrome has been reported. Facial clefting can be identified, although demonstration of isolated cleft palate is difficult (Twining 2007). Abnormalities of the fetal neck are uncommon.

POSTNATAL GROWTH OF THE SKULL

Postnatal growth of the skull is characterized by changing proportions of its components. Growth of the brain continues to be extremely rapid in the first 2 years. The metopic suture fuses during the first year, by which time the rapid phase of increase in breadth of the forehead is complete; further growth and remodelling of the frontal bones is mediated by appositional growth. The sagittal suture continues as an active growth centre until puberty, when growth of the brain is complete. In addition to continuing growth of the frontal and parietal bones, the squamous temporal bone increases in size so that it contributes a greater proportion of the calvaria in the adult than in the neonate (Fig. 36.22). The inner ear and the petrous temporal bone around it grow very little after birth, so the increasing breadth of the skull draws the petrous temporal bone out laterally, creating the bony external acoustic meatus. The tympanic ring (with the tympanic membrane) lies at the surface of the meatus in the neonatal skull; it remains at the proximal end of the deepening canal. Use of sternocleidomastoid to lift the head results in formation of the mastoid process of the temporal bone, which develops air-filled spaces (mastoid air cells) that are continuous with the middle ear cavity. The paranasal sinuses begin to form in late fetal life as diverticula from the nasal cavity that gradually invade the maxilla, frontal, ethmoid and sphenoid bones. At birth, small ethmoidal and maxillary sinuses are present, but the frontal sinus is nothing more than an out-pouching from the nasal cavity, and there is no pneumatization of the sphenoid bone (p. 566). Thickening of the skull bones is accompanied by increasing size of the sinuses, and by a change in form of the sutures, from straight to wavy lines and, finally, to the complex interdigitations seen in the adult. After growth ceases, the skull sutures contain inert connective tissue and some cartilage; in old age, some of them are completely replaced by bone (natural synostosis).

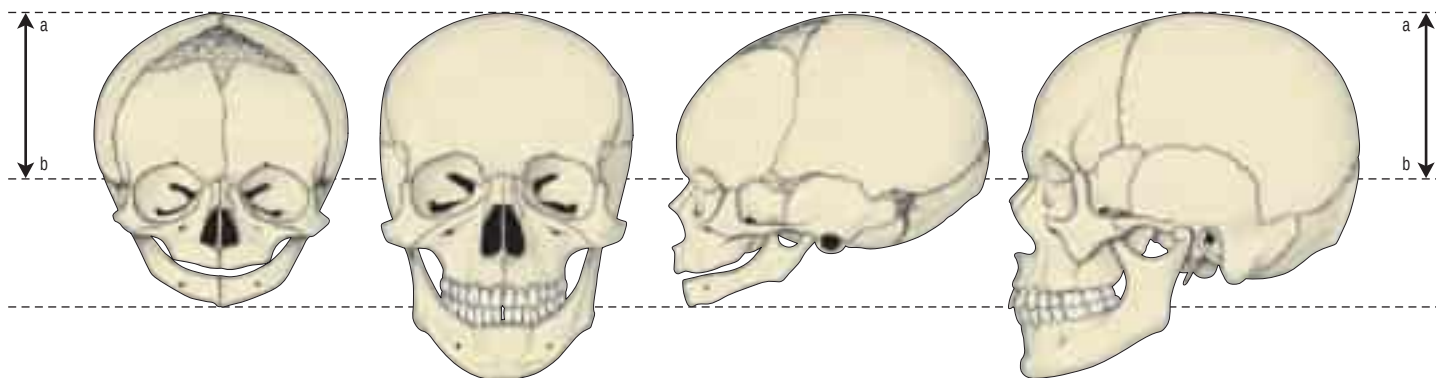


Fig. 36.22 During postnatal growth of the skull, the viscerocranium increases in size relative to the neurocranium. This diagram shows that, with the height of the cranial vault expressed as similar in newborn and adult skulls (lines a↔b), the facial skeleton increases particularly rapidly during childhood and puberty.

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